10

Rotation of a Rigid Object About a Fixed Axis

CHAPTER OUTLINE

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* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ10.1 Answer (c). The wheel has a radius of 0.500 m and made 320 revolutions. The distance traveled is

$$s = r\theta = (0.500\text{ m})(320\text{ rev})\left(\frac{2\pi\text{ rad}}{1\text{ rev}}\right) = 1.00\times10^{3}\text{ m} = 1.00\text{ km}$$

OQ10.2 Answer (b). Any object moving in a circular path undergoes a constant change in the direction of its velocity. This change in the direction of velocity is an acceleration, always directed toward the center of the path, called the centripetal acceleration, $a_c = v^2/r = r\omega^2$. The tangential speed of the object is $v_t = r\omega$, where ω is the angular velocity. If ω is not constant, the object will have both an angular

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acceleration, $\alpha_{\text{avg}} = \Delta\omega / \Delta t$, and a tangential acceleration, $a_t = r\alpha$. The only untrue statement among the listed choices is (b). Even when ω is constant, the object still has centripetal acceleration.

OQ10.3 Answer: b = e > a = d > c = 0. The tangential acceleration has magnitude $(3/\text{s}^2)r$, where r is the radius. It is constant in time. The radial acceleration has magnitude $\omega^2 r$, so it is $(4/\text{s}^2)r$ at the first and last moments mentioned and it is zero at the moment the wheel reverses.

OQ10.4 Answer (d). The angular displacement will be

$$\Delta\theta = \omega_{\text{avg}}\Delta t = \left(\frac{\omega_f + \omega_i}{2}\right)\Delta t$$

$$= \left(\frac{12.00\ \text{rad/s} + 4.00\ \text{rad/s}}{2}\right)(4.00\ \text{s}) = 32.0\ \text{rad}$$

OQ10.5 **(i)** Answer (d). The speedometer measures the number of revolutions per second of the tires. A larger tire will rotate fewer times to cover the same distance. The speedometer reading is assumed proportional to the rotation rate of the tires, $\omega = v/R$, for a standard tire radius R, but the actual reading is $\omega = v/(1.3)R$, or 1.3 times smaller. Example: When the car travels at 13 km/h, the speedometer reads 10 km/h.

(ii) Answer (d). If the driver uses the odometer reading to calculate fuel economy, this reading is a factor of 1.3 too small because the odometer assumes 1 rev = $2\pi R$ for a standard tire radius R, whereas the actual distance traveled is $1.3(2\pi R)$, so the fuel economy in miles per gallon will appear to be lower by a factor of 1.3. Example: If the car travels 13 km, the odometer will read 10 km. If the car actually makes 13 km/gal, the calculation will give 10 km/gal.

OQ10.6 **(i)** Answer (a). Smallest I is about the x axis, along which the larger-mass balls lie.

(ii) Answer (c). The balls all lie at a distance from the z axis, which is perpendicular to both the x and y axes and passes through the origin.

OQ10.7 Answer (a). The accelerations are not equal, but greater in case (a). The string tension above the 50-N object is less than its weight while the object is accelerating downward because it does not fall with the acceleration of gravity.

OQ10.8 Answers (a), (b), (e). The object must rotate with a nonzero and constant angular acceleration. Its moment of inertia would not

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change unless there were a rearrangement of mass within the object.

OQ10.9 (**i**) Answer (a). The basketball has rotational as well as translational kinetic energy.

(**ii**) Answer (c). The motions of their centers of mass are identical.

(**iii**) Answer (a). The basketball-Earth system has more kinetic energy than the ice-Earth system due to the rotational kinetic energy of the basketball. Therefore, when the kinetic energy of both systems has transformed to gravitational potential energy when the objects momentarily come to rest at their highest point on the ramp, the basketball will be at a higher location, corresponding to the larger gravitational potential energy.

OQ10.10 **(i)** Answer (c). The airplane momentarily has zero torque acting on it. It was speeding up in its angular rotation before this instant of time and begins slowing down just after this instant.

(**ii**) Answer (b). Although the angular speed is zero at this instant, there is still an angular acceleration because the wound-up string applies a torque to the airplane. This is similar to a ball thrown upward, which we studied earlier: at the top of its flight, it momentarily comes to rest, but is still accelerating because the gravitational force is acting on it.

OQ10.11 Answer (e). The sphere of twice the radius has eight times the volume and eight times the mass, and the r^2 term in $I = \frac{2}{5}mr^2$ also becomes four times larger.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ10.1 Yes. For any object on which a net force acts but no net torque, the translational kinetic energy will change but the rotational kinetic energy will not. For example, if you drop an object, it will gain translational kinetic energy due to work done on the object by the gravitational force. Any rotational kinetic energy the object has is unaffected by dropping it.

CQ10.2 No, just as an object need not be moving to have mass.

CQ10.3 If the object is free to rotate about any axis, the object will start to rotate if the two forces act along <u>different</u> lines of action. Then the torques of the forces will not be equal in magnitude and opposite in direction.

CQ10.4 Attach an object, of known mass m, to the cord. You could measure

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the time that it takes the object to fall a measured distance after being released from rest. Using this information, the linear acceleration of the mass can be calculated, and then the torque on the rotating object and its angular acceleration. It is assumed the mass of the cord has negligible effect on the motion as the cord unwinds.

CQ10.5 We have from Example 10.6 the means to calculate a and α. You could use $\omega = \alpha t$ and $v = at$.

CQ10.6 The moment of inertia depends on the distribution of mass with respect to a given axis. If the axis is changed, then each bit of mass that makes up the object is at a different distance from the axis than before. Compare the moments of inertia of a uniform rigid rod about axes perpendicular to the rod, first passing through its center of mass, then passing through an end. For example, if you wiggle repeatedly a meterstick back and forth about an axis passing through its center of mass, you will find it does not take much effort to reverse the direction of rotation. However, if you move the axis to an end, you will find it more difficult to wiggle the stick back and forth. The moment of inertia about the end is much larger, because much of the mass of the stick is farther from the axis.

CQ10.11 No, only if its angular velocity changes.

CQ10.12 Adding a small sphere of mass m to the end will increase the moment of inertia of the system from $(1/3)ML^2$ to $(1/3)ML^2 + mL^2$, and the initial potential energy would be $(1/2)MgL + mgL$. Following Example 10.11, the final angular speed ω would be

$$\omega = \sqrt{\frac{3g}{L}}\sqrt{\frac{M+2m}{M+3m}}$$

If $m = M$, $\omega = \sqrt{\frac{3g}{L}}\sqrt{\frac{M+2m}{M+3m}} = \sqrt{\frac{3g}{L}}\sqrt{\frac{3M}{4M}} = \sqrt{\frac{9g}{4L}}$

Therefore, ω would increase.

CQ10.13 (a) The sphere would reach the bottom first. (b) The hollow cylinder would reach the bottom last. First imagine that each object has the same mass and the same radius. Then they all have the same torque due to gravity acting on them. The one with the smallest moment of inertia will thus have the largest angular acceleration and reach the bottom of the plane first. Equation 10.52 describes the speed of an object rolling down an inclined plane. In the denominator, I_{CM} will be a numerical factor (e.g., 2/5 for the sphere) multiplied by MR^2. Therefore, the mass and radius will cancel in the equation and the center-of-mass speed will be independent of mass and radius.

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CQ10.14 (a) Sewer pipe: $I_{CM} = MR^2$. (b) Embroidery hoop: $I_{CM} = MR^2$. (c) Door: $I = \frac{1}{3}MR^2$. (d) Coin: $I_{CM} = \frac{1}{2}MR^2$. The distribution of mass along lines parallel to the axis makes no difference to the moment of inertia.

CQ10.15 (a) The tricycle rolls forward. (b) The tricycle rolls forward. (c) The tricycle rolls backward. (d) The tricycle does not roll, but may skid forward. (e) The tricycle rolls backward. (f) To answer these questions, think about the torque of the string tension about an axis at the bottom of the wheel, where the rubber meets the road. This is the instantaneous axis of rotation in rolling. Cords A and B produce clockwise torques about this axis. Cords C and E produce counterclockwise torques. Cord D has zero lever arm.

CQ10.16 As one finger slides towards the center, the normal force exerted by the sliding finger on the ruler increases. At some point, this normal force will increase enough so that static friction between the sliding finger and the ruler will stop their relative motion. At this moment the other finger starts sliding along the ruler towards the center. This process repeats until the fingers meet at the center of the ruler.

Next step: Try a rod with a nonuniform mass distribution.

Next step: Wear a piece of sandpaper as a ring on one finger to change its coefficient of friction.

SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 10.1 Angular Position, Velocity, and Acceleration

P10.1 (a) The Earth rotates 2π radians (360°) on its axis in 1 day. Thus,

$$\omega = \frac{\Delta\theta}{\Delta t} = \frac{2\pi \text{ rad}}{1 \text{ day}}\left(\frac{1 \text{ day}}{8.64\times10^4 \text{ s}}\right) = \boxed{7.27\times10^{-5} \text{ rad/s}}$$

(b) Because of its angular speed, the Earth bulges at the equator.

P10.2 (a) $$\alpha = \frac{\Delta\omega}{\Delta t} = \frac{1.00 \text{ rev/s} - 0}{30.0 \text{ s}} = \left(3.33\times10^{-2} \frac{\text{rev}}{\text{s}^2}\right)\left(\frac{2\pi \text{ rad}}{1 \text{ rev}}\right)$$

$$= \boxed{0.209 \text{ rad/s}^2}$$

(b) Yes. When an object starts from rest, its angular speed is related to the angular acceleration and time by the equation $\omega = \alpha(\Delta t)$.

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Thus, the angular speed is directly proportional to both the angular acceleration and the time interval. If the time interval is held constant, doubling the angular acceleration will double the angular speed attained during the interval.

P10.3 (a) $\theta|_{t=0} = \boxed{5.00 \text{ rad}}$

$$\omega|_{t=0} = \left.\frac{d\theta}{dt}\right|_{t=0} = 10.0 + 4.00t|_{t=0} = \boxed{10.0 \text{ rad/s}}$$

$$\alpha|_{t=0} = \left.\frac{d\omega}{dt}\right|_{t=0} = \boxed{4.00 \text{ rad/s}^2}$$

(b) $\theta|_{t=3.00\text{ s}} = 5.00 + 30.0 + 18.0 = \boxed{53.0 \text{ rad}}$

$$\omega|_{t=3.00s} = \left.\frac{d\theta}{dt}\right|_{t=3.00s} = 10.0 + 4.00t|_{t=3.00s} = \boxed{22.0 \text{ rad/s}}$$

$$\alpha|_{t=3.00s} = \left.\frac{d\omega}{dt}\right|_{t=3.00s} = \boxed{4.00 \text{ rad/s}^2}$$

P10.4 $$\alpha = \frac{d\omega}{dt} = 10 + 6t \quad \rightarrow \quad \int_0^\omega d\omega = \int_0^t (10 + 6t)\,dt \quad \rightarrow \quad \omega - 0 = 10t + \frac{6}{2}t^2$$

$$\omega = \frac{d\theta}{dt} = 10t + 3t^2 \quad \rightarrow \quad \int_0^\theta d\theta = \int_0^t (10t + 3t^2)\,dt \quad \rightarrow \quad \theta - 0 = \frac{10t^2}{2} + \frac{3t^3}{3}$$

$\theta = 5t^2 + t^3$. At $t = 4.00$ s, $\theta = 5(4.00 \text{ s})^2 + (4.00 \text{ s})^3 = \boxed{144 \text{ rad}}$

Section 10.2 Analysis Model: Rigid Object Under Constant Angular Acceleration

P10.5 (a) We start with $\omega_f = \omega_i + \alpha t$ and solve for the angular acceleration α:

$$\alpha = \frac{\omega - \omega_i}{t} = \frac{12.0 \text{ rad/s}}{3.00 \text{ s}} = \boxed{4.00 \text{ rad/s}^2}$$

(b) The angular position of a rigid object under constant angular acceleration is given by Equation 10.7:

$$\theta = \omega_i t + \frac{1}{2}\alpha t^2 = \frac{1}{2}(4.00 \text{ rad/s}^2)(3.00 \text{ s})^2 = \boxed{18.0 \text{ rad}}$$

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P10.6 $\omega_i = 3\,600 \text{ rev/min} = 3.77 \times 10^2 \text{ rad/s}$

$\theta = 50.0 \text{ rev} = 3.14 \times 10^2 \text{ rad}$ and $\omega_f = 0$

$$\omega_f^2 = \omega_i^2 + 2\alpha\theta$$

$$0 = \left(3.77 \times 10^2 \text{ rad/s}\right)^2 + 2\alpha\left(3.14 \times 10^2 \text{ rad}\right)$$

$$\alpha = \boxed{-2.26 \times 10^2 \text{ rad/s}^2}$$

P10.7 We are given $\alpha = -2.00 \text{ rad/s}^2$, $\omega_f = 0$, and

$$\omega_i = \frac{100 \text{ rev}}{1.00 \text{ min}}\left(\frac{1 \text{ min}}{60.0 \text{ s}}\right)\left(\frac{2\pi \text{ rad}}{1.00 \text{ rev}}\right) = \frac{10\pi}{3} \text{ rad/s}$$

(a) From $\omega_f - \omega_i = \alpha t$, we have

$$t = \frac{\omega_f - \omega_i}{\alpha} = \frac{0 - (10\pi / 3)}{-2.00} \text{ s} = \boxed{5.24 \text{ s}}$$

(b) Since the motion occurs with constant angular acceleration, we write

$$\theta_f = \bar{\omega} t = \left(\frac{\omega_f + \omega_i}{2}\right) t = \left(\frac{10\pi}{6} \text{ rad/s}\right)\left(\frac{10\pi}{6} \text{ s}\right) = \boxed{27.4 \text{ rad}}$$

P10.8 (a) From $\omega_f^2 = \omega_i^2 + 2\alpha(\Delta\theta)$, the angular displacement is

$$\Delta\theta = \frac{\omega_f^2 - \omega_i^2}{2\alpha} = \frac{(2.2 \text{ rad/s})^2 - (0.06 \text{ rad/s})^2}{2(0.70 \text{ rad/s}^2)} = \boxed{3.5 \text{ rad}}$$

(b) From the equation given above for $\Delta\theta$, observe that when the angular acceleration is constant, the displacement is proportional to the difference in the *squares* of the final and initial angular speeds. Thus, the angular displacement would $\boxed{\text{increase by a factor of 4}}$ if both of these speeds were doubled.

***P10.9** We are given $\omega_f = 2.51 \times 10^4 \text{ rev/min} = 2.63 \times 10^3 \text{ rad/s}$

(a) $$\alpha = \frac{\omega_f - \omega_i}{t} = \frac{2.63 \times 10^3 \text{ rad/s} - 0}{3.20 \text{ s}} = \boxed{8.21 \times 10^2 \text{ rad/s}^2}$$

(b) $$\theta_f = \omega_i t + \frac{1}{2}\alpha t^2 = 0 + \frac{1}{2}(8.21 \times 10^2 \text{ rad/s}^2)(3.20 \text{ s})^2 = \boxed{4.21 \times 10^3 \text{ rad}}$$

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P10.10 According to the definition of average angular speed (Eq. 10.2), the disk's average angular speed is 50.0 rad/10.0 s = 5.00 rad/s. According to the average angular speed expressed as $(\omega_i + \omega_f)/2$ in the model of a rigid object under constant angular acceleration, the average angular speed of the disk is (0 + 8.00 rad/s)/2 = 4.00 rad/s. Because these two numbers do not match, the angular acceleration of the disk cannot be constant.

P10.11 $\theta_f - \theta_i = \omega_i t + \frac{1}{2}\alpha t^2$ and $\omega_f = \omega_i + \alpha t$ are two equations in two unknowns, ω_i and α.

$$\omega_i = \omega_f - \alpha t\text{: } \theta_f - \theta_i = \left(\omega_f - \alpha t\right)t + \frac{1}{2}\alpha t^2 = \omega t - \frac{1}{2}\alpha t^2$$

$$\left(37.0 \text{ rev}\right)\left(\frac{2\pi \text{ rad}}{1 \text{ rev}}\right) = \left(98.0 \text{ rad/s}\right)\left(3.00 \text{ s}\right) - \frac{1}{2}\alpha\left(3.00 \text{ s}\right)^2$$

$$232 \text{ rad} = 294 \text{ rad} - \left(4.50 \text{ s}^2\right)\alpha\text{: } \quad \alpha = \frac{61.5 \text{ rad}}{4.50 \text{ s}^2} = \boxed{13.7 \text{ rad/s}^2}$$

P10.12 $\omega = 5.00 \text{ rev/s} = 10.0\pi \text{ rad/s}$. We will break the motion into two stages: (1) a period during which the tub speeds up and (2) a period during which it slows down.

While speeding up, $\theta_1 = \bar{\omega}t = \frac{0 + 10.0\pi \text{ rad/s}}{2}(8.00 \text{ s}) = 40.0\pi \text{ rad}$.

While slowing down, $\theta_2 = \bar{\omega}t = \frac{10.0\pi \text{ rad/s} + 0}{2}(12.0 \text{ s}) = 60.0\pi \text{ rad}$.

So, $\theta_{\text{total}} = \theta_1 + \theta_2 = 100\pi \text{ rad} = \boxed{50.0 \text{ rev}}$.

***P10.13** We use $\theta_f - \theta_i = \omega_i t + \frac{1}{2}\alpha t^2$ and $\omega_f = \omega_i + \alpha t$ to obtain

$$\omega_i = \omega_f - \alpha t \quad \text{and} \quad \theta_f - \theta_i = \left(\omega_f - \alpha t\right)t + \frac{1}{2}\alpha t^2 = \omega_f t - \frac{1}{2}\alpha t^2$$

Solving for the final angular speed gives

$$\omega_f = \frac{\theta_f - \theta_i}{t} + \frac{1}{2}\alpha t = \frac{62.4 \text{ rad}}{4.20 \text{ s}} + \frac{1}{2}\left(-5.60 \text{ rad/s}^2\right)(4.20 \text{ s})$$

$$= \boxed{3.10 \text{ rad/s}^2}$$

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P10.14 (a) Let R_E represent the radius of the Earth. The base of the building moves east at $v_1 = \omega R_E$, where ω is one revolution per day. The top of the building moves east at $v_2 = \omega(R_E + h)$. Its eastward speed relative to the ground is $v_2 - v_1 = \omega h$. The object's time of fall is given by $\Delta y = 0 + \frac{1}{2}gt^2$, $t = \sqrt{\frac{2h}{g}}$. During its fall the object's eastward motion is unimpeded so its deflection distance is

$$\Delta x = (v_2 - v_1)t = \omega h\sqrt{\frac{2h}{g}} = \boxed{\omega h^{3/2}\left(\frac{2}{g}\right)^{1/2}}$$

(b) $$\left(\frac{2\pi \text{ rad}}{86400 \text{ s}}\right)(50.0 \text{ m})^{3/2}\left(\frac{2}{9.80 \text{ m/s}^2}\right)^{1/2} = \boxed{1.16 \text{ cm}}$$

(c) The deflection is only 0.02% of the original height, so it is negligible in many practical cases.

(d) Decrease. Because the displacement is proportional to angular speed and the angular acceleration is constant, the displacement decreases linearly in time.

Section 10.3 Angular and Translational Quantities

P10.15 (a) From $v = r\omega$, we have

$$\omega = \frac{v}{r} = \frac{45.0 \text{ m/s}}{250 \text{ m}} = \boxed{0.180 \text{ rad/s}}$$

(b) Traveling at constant speed along a circular track, the car will experience a centripetal acceleration given by

$$a_r = \frac{v^2}{r} = \frac{(45.0 \text{ m/s})^2}{250 \text{ m}} = \boxed{8.10 \text{ m/s}^2 \text{ toward the center of track}}$$

P10.16 Estimate the tire's radius at 0.250 m and miles driven as 10 000 per year. Then,

$$\theta = \frac{s}{r} = \left(\frac{1.00 \times 10^4 \text{ mi}}{0.250 \text{ m}}\right)\left(\frac{1609 \text{ m}}{1 \text{ mi}}\right) = 6.44 \times 10^7 \text{ rad/yr}$$

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$$\theta = \left(6.44\times10^{7}\ \text{rad/yr}\right)\left(\frac{1\,\text{rev}}{2\pi\ \text{rad}}\right) = 1.02\times10^{7}\ \text{rev/yr or } \boxed{\sim 10^{7}\ \text{rev/yr}}$$

P10.17 (a) The final angular speed is

$$\omega = \frac{v}{r} = \frac{25.0\ \text{m/s}}{1.00\ \text{m}} = \boxed{25.0\ \text{rad/s}}$$

(b) We solve for the angular acceleration from $\omega_f^2 = \omega_i^2 + 2\alpha(\Delta\theta)$:

$$\alpha = \frac{\omega_f^2 - \omega_i^2}{2(\Delta\theta)} = \frac{(25.0\ \text{rad/s})^2 - 0}{2\left[(1.25\ \text{rev})(2\pi\ \text{rad/rev})\right]} = \boxed{39.8\ \text{rad/s}^2}$$

(c) From the definition of angular acceleration,

$$\Delta t = \frac{\Delta\omega}{\alpha} = \frac{25.0\ \text{rad/s}}{39.8\ \text{rad/s}^2} = \boxed{0.628\ \text{s}}$$

P10.18 (a) Consider a tooth on the front sprocket. It gives this speed, relative to the frame, to the link of the chain it engages:

$$v = r\omega = \left(\frac{0.152\ \text{m}}{2}\right)(76\ \text{rev/min})\left(\frac{2\pi\ \text{rad}}{1\ \text{rev}}\right)\left(\frac{1\ \text{min}}{60\ \text{s}}\right)$$

$$= \boxed{0.605\ \text{m/s}}$$

(b) Consider the chain link engaging a tooth on the rear sprocket:

$$\omega = \frac{v}{r} = \frac{0.605\ \text{m/s}}{(0.070\ \text{m})/2} = \boxed{17.3\ \text{rad/s}}$$

(c) Consider the wheel tread and the road. A thread could be unwinding from the tire with this speed relative to the frame:

$$v = r\omega = \left(\frac{0.673\ \text{m}}{2}\right)(17.3\ \text{rad/s}) = \boxed{5.82\ \text{m/s}}$$

(d) $\boxed{\text{We did not need to know the length of the pedal cranks}}$, but we could use that information to find the linear speed of the pedals:

$$v = r\omega = (0.175\ \text{m})(7.96\ \text{rad/s})\left(\frac{1}{1\ \text{rad}}\right) = 1.39\ \text{m/s}$$

P10.19 Given $r = 1.00$ m, $\alpha = 4.00\ \text{rad/s}^2$, $\omega_i = 0$, and $\theta_i = 57.3° = 1.00$ rad:

(a) $\omega_f = \omega_i + \alpha t = 0 + \alpha t$

At $t = 2.00$ s, $\omega_f = 4.00\ \text{rad/s}^2\,(2.00\ \text{s}) = \boxed{8.00\ \text{rad/s}}$

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(b) $v = r\omega = (1.00\text{ m})(8.00\text{ rad/s}) = \boxed{8.00\text{ m/s}}$

(c) $|a_r| = a_c = r\omega^2 = (1.00\text{ m})(8.00\text{ rad/s})^2 = 64.0\text{ m/s}^2$

$a_t = r\alpha = (1.00\text{ m})(4.00\text{ rad/s}^2) = 4.00\text{ m/s}^2$

The magnitude of the total acceleration is

$$a = \sqrt{a_r^2 + a_t^2} = \sqrt{(64.0\text{ m/s}^2)^2 + (4.00\text{ m/s}^2)^2} = \boxed{64.1\text{ m/s}^2}$$

The direction the total acceleration vector makes with respect to the radius to point P is

$$\phi = \tan^{-1}\left(\frac{a_t}{a_c}\right) = \tan^{-1}\left(\frac{4.00}{64.0}\right) = \boxed{3.58°}$$

(d) $\theta_f = \theta_i + \omega_i t + \frac{1}{2}\alpha t^2 = (1.00\text{ rad}) + \frac{1}{2}(4.00\text{ rad/s}^2)(2.00\text{ s})^2 = \boxed{9.00\text{ rad}}$

P10.20 (a) We first determine the distance travelled by the car during the 9.00-s interval:

$$s = \bar{v}t = \frac{v_i + v_f}{2}t = (11.0\text{ m/s})(9.00\text{ s}) = 99.0\text{ m}$$

the number of revolutions completed by the tire is then

$$\theta = \frac{s}{r} = \frac{99.0\text{ m}}{0.290\text{ m}} = 341\text{ rad} = \boxed{54.3\text{ rev}}$$

(b) $\omega_f = \frac{v_f}{r} = \frac{22.0\text{ m/s}}{0.290\text{ m}} = 75.9\text{ rad/s} = \boxed{12.1\text{ rev/s}}$

P10.21 Every part of this problem is about using radian measure to relate rotation of the whole object to the linear motion of a point on the object.

(a) $\omega = 2\pi f = \frac{2\pi\text{ rad}}{1\text{ rev}}\left(\frac{1\,200\text{ rev}}{60.0\text{ s}}\right) = \boxed{126\text{ rad/s}}$

(b) $v = \omega r = (126\text{ rad/s})(3.00\times10^{-2}\text{ m}) = \boxed{3.77\text{ m/s}}$

(c) $a_c = \omega^2 r = (126\text{ rad/s})^2(8.00\times10^{-2}\text{ m}) = 1\,260\text{ m/s}^2$ so

$\vec{\mathbf{a}}_r = \boxed{1.26\text{ km/s}^2\text{ toward the center}}$

(d) $s = r\theta = \omega r t = (126\text{ rad/s})(8.00\times10^{-2}\text{ m})(2.00\text{ s}) = \boxed{20.1\text{ m}}$

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P10.22 (a) $\boxed{5.77 \text{ cm}}$

(b) Yes. The top of the ladder is displaced

$$\theta = s/r = 0.690\,\text{m}/4.90\,\text{m} \cong 0.141 \text{ rad}$$

from vertical about its right foot. The left foot of the ladder is displaced by the same angle below the horizontal; therefore,

$$\theta = 0.690\,\text{m}/4.90\,\text{m} = t/0.410\,\text{m} \rightarrow t = 5.77 \text{ cm}$$

Note that we are approximating the straight-line distance of 0.690 m as an arc length because it is much smaller than the length of the ladder. The thickness of the rock is a cruder approximation of an arc length because the rung of the ladder is much shorter than the length of the ladder.

P10.23 The force of static friction must act forward and then more and more inward on the tires, to produce both tangential and centripetal acceleration. Its tangential component is $m\left(1.70\ \text{m/s}^2\right)$. Its radially inward component is $ma_c = \dfrac{mv^2}{r} = m\omega^2 r,$ which increases with time: this takes the maximum value

$$m\omega_f^2 r = mr\left(\omega_i^2 + 2\alpha\Delta\theta\right) = mr\left(0 + 2\alpha\frac{\pi}{2}\right) = m\pi r\alpha = m\pi a_t$$

$$= m\pi\left(1.70\,\text{m/s}^2\right)$$

With skidding impending we have $\sum F_y = ma_y$, $+n - mg = 0$, $n = mg$:

$$f_s = \mu_s n = \mu_s mg = \sqrt{m^2\left(1.70\ \text{m/s}^2\right)^2 + m^2\pi^2\left(1.70\ \text{m/s}^2\right)^2}$$

$$\mu_s = \frac{1.70\ \text{m/s}^2}{g}\sqrt{1+\pi^2} = \boxed{0.572}$$

P10.24 The force of static friction must act forward and then more and more inward on the tires, to produce both tangential and centripetal acceleration. Its tangential component is $ma = mr\alpha.$ Its radially inward component is $ma_c = \dfrac{mv^2}{r} = m\omega^2 r$ which increases with time; this takes the maximum value

$$m\omega_f^2 r = mr\left(\omega_i^2 + 2\alpha\Delta\theta\right) = mr\left(0 + 2\alpha\frac{\pi}{2}\right) = m\pi r\alpha = m\pi a$$

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With skidding impending we have

$$\sum F_y = ma_y: \quad +n - mg = 0 \quad \rightarrow \quad n = mg$$

$$f_s = \mu_s n = \mu_s mg = \sqrt{\left(ma_t\right)^2 + \left(ma_c\right)^2} = \sqrt{m^2a^2 + m^2\pi^2a^2}$$

$$\mu_s = \boxed{\frac{a}{g}\sqrt{1+\pi^2}}$$

P10.25 (a) The general expression for angular velocity is

$$\omega = \frac{d\theta}{dt} = \frac{d}{dt}\left(2.50t^2 - 0.600t^3\right) = 5.00t - 1.80t^2$$

where ω is in radians/second and t is in seconds.

The angular velocity will be a maximum when

$$\frac{d\omega}{dt} = \frac{d}{dt}\left(5.00t - 1.80t^2\right) = 5.00 - 3.60t = 0$$

Solving for the time t, we find

$$t = \frac{5.00}{3.60} = 1.39\,\text{s}$$

Placing this value for t into the equation for angular velocity, we find

$$\omega_{\max} = 5.00t - 1.80t^2 = 5.00\left(1.39\right) - 1.80\left(1.39\right)^2 = \boxed{3.47\,\text{rad/s}}$$

(b) $v_{\max} = \omega_{\max} r = (3.47\,\text{rad/s})(0.500\text{ m}) = \boxed{1.74\,\text{m/s}}$

(c) The roller reverses its direction when the angular velocity is zero—recall an object moving vertically upward against gravity reverses its motion when its velocity reaches zero at the maximum height.

$$\omega = 5.00t - 1.80t^2 = t(5.00 - 1.80t) = 0$$

$$\rightarrow 5.00 - 1.80t = 0 \rightarrow t = \frac{5.00}{1.80} = 2.78\,\text{s}$$

The driving force should be removed from the roller at $t = \boxed{2.78\,\text{s}}$.

(d) Set t = 2.78 s in the expression for angular position:

$$\theta = 2.50t^2 - 0.600t^3 = 2.50\left(2.78\right)^2 - 0.600\left(2.78\right)^3 = 6.43\,\text{rad}$$

or $$(6.43\text{ rad})\left(\frac{1\text{ rotation}}{2\pi\,\text{rad}}\right) = \boxed{1.02\text{ rotations}}$$

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P10.26 The object starts with $\theta_i = 0$. The location of its final position on the circle is found from $9\,\text{rad} - 2\pi = 2.72\,\text{rad} = 156°$.

(a) Its position vector is

$$3.00\text{ m at }156° = (3.00\text{ m})\cos 156°\,\hat{\mathbf{i}} + (3.00\text{ m})\sin 156°\,\hat{\mathbf{j}}$$

$$= \boxed{\left(-2.73\hat{\mathbf{i}} + 1.24\hat{\mathbf{j}}\right)\text{ m}}$$

(b) $\boxed{\text{It is in the second quadrant, at } 156°}$

(c) The object's velocity is $v = \omega r = (1.50\text{ rad/s})(3.00\text{ m}) = 4.50\text{ m/s}$ at 90°. After the displacement, its velocity is

$$4.50\text{ m/s at } 90° + 156° \quad \text{or}$$

$$4.50\text{ m/s at } 246° = (4.50\text{ m/s})\cos 246°\,\hat{\mathbf{i}} + (4.50\text{ m/s})\sin 246°\,\hat{\mathbf{j}}$$

$$= \boxed{\left(-1.85\hat{\mathbf{i}} - 4.10\hat{\mathbf{j}}\right)\text{ m/s}}$$

(d) $\boxed{\text{It is moving toward the third quadrant, at } 246°}$.

(e) Its acceleration is v^2/r, opposite in direction to its position vector. This is

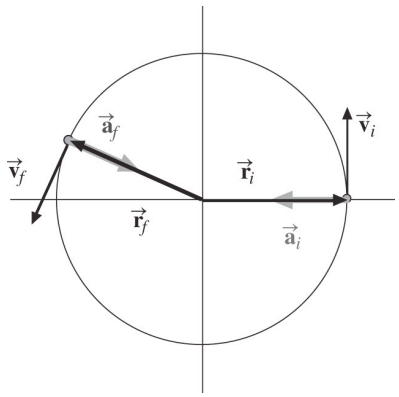
$$\frac{(4.50\text{ m/s})^2}{3.00\text{ m}} \text{ at } \quad 180° + 156° \quad \text{or}$$

$$6.75\text{ m/s}^2 \text{ at } 336° = \left(6.75\text{ m/s}^2\right)\cos 336°\,\hat{\mathbf{i}}$$

$$+ \left(6.75\text{ m/s}^2\right)\sin 336°\,\hat{\mathbf{j}}$$

$$= \boxed{\left(6.15\hat{\mathbf{i}} - 2.78\hat{\mathbf{j}}\right)\text{ m/s}^2}$$

(f) ANS. FIG. P10.26 shows the initial and final position, velocity, and acceleration vectors.



ANS. FIG. P10.26

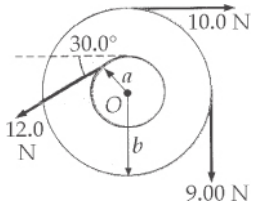
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(g) The total force is given by

$$F = ma = (4.00\ \text{kg})(6.15\hat{\mathbf{i}} - 2.78\ \hat{\mathbf{j}})\ \text{m/s}^2 = \boxed{(24.6\ \hat{\mathbf{i}} - 11.1\ \hat{\mathbf{j}})\ \text{N}}$$

Section 10.4 Torque

P10.27 To find the net torque, we add the individual torques, remembering to apply the convention that a torque producing clockwise rotation is negative and a counterclockwise rotation is positive.



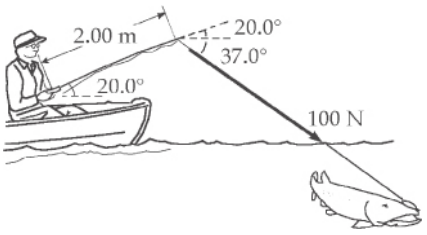
ANS. FIG. P10.27

$$\sum \tau = (0.100\ \text{m})(12.0\ \text{N}) - (0.250\ \text{m})(9.00\ \text{N}) - (0.250\ \text{m})(10.0\ \text{N})$$

$$= \boxed{-3.55\ \text{N}\cdot\text{m}}$$

The thirty-degree angle is unnecessary information.

P10.28 We resolve the 100-N force into components perpendicular to and parallel to the rod, as



ANS. FIG. P10.28

$$F_{\text{par}} = (100\ \text{N})\cos 57.0° = 54.5\ \text{N}$$

and

$$F_{\text{perp}} = (100\ \text{N})\sin 57.0° = 83.9\ \text{N}$$

The torque of F_{par} is zero since its line of action passes through the pivot point.

The torque of F_{perp} is

$$\tau = (83.9\ \text{N})(2.00\ \text{m}) = \boxed{168\ \text{N}\cdot\text{m}}\ \text{(clockwise)}$$

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Section 10.5 Analysis Model: Rigid Object Under a Net Torque

P10.29 The flywheel is a solid disk of mass M and radius R with axis through its center.

$$\left.\begin{array}{l}\sum\tau = I\alpha \\ I = \dfrac{1}{2}MR^2\end{array}\right\} -T_u r + T_b r = \frac{1}{2}MR^2\alpha \rightarrow T_b = T_u + \frac{MR^2\alpha}{2r}$$

$$T_b = 135\text{ N} + \frac{(80.0\text{ kg})(0.625\text{ m})^2(-1.67\text{ rad/s}^2)}{2(0.230\text{ m})} = \boxed{21.5\text{ N}}$$

P10.30 (a) The moment of inertia of the wheel, modeled as a disk, is

$$I = \frac{1}{2}MR^2 = \frac{1}{2}(2.00\text{ kg})(7.00\times10^{-2}\text{ m})^2 = 4.90\times10^{-3}\text{ kg}\cdot\text{m}^2$$

From Newton's second law for rotational motion,

$$\alpha = \frac{\sum\tau}{I} = \frac{0.600}{4.90\times10^{-3}} = 122\text{ rad/s}^2$$

then, from $\alpha = \dfrac{\Delta\omega}{\Delta t}$, we obtain

$$\Delta t = \frac{\Delta\omega}{\alpha} = \frac{1200(2\pi/60)}{122} = \boxed{1.03\text{ s}}$$

(b) The number of revolutions is determined from

$$\Delta\theta = \frac{1}{2}\alpha t^2 = \frac{1}{2}(122\text{ rad/s})(1.03\text{ s})^2 = 64.7\text{ rad} = \boxed{10.3\text{ rev}}$$

***P10.31** (a) We first determine the moment of inertia of the merry-go-round:

$$I = \frac{1}{2}MR^2 = \frac{1}{2}(150\text{ kg})(1.50\text{ m})^2 = 169\text{ kg}\cdot\text{m}^2$$

To find the angular acceleration, we use

$$\alpha = \frac{\Delta\omega}{\Delta t} = \frac{\omega_f - \omega_i}{\Delta t} = \left(\frac{0.500\text{ rev/s} - 0}{2.00\text{ s}}\right)\left(\frac{2\pi\text{ rad}}{1\text{ rev}}\right) = \frac{\pi}{2}\text{ rad/s}^2$$

From the definition of torque, $\tau = F\cdot r = I\alpha$, we obtain

$$F = \frac{I\alpha}{r} = \frac{(169\text{ kg}\cdot\text{m}^2)\left(\frac{\pi}{2}\text{ rad/s}^2\right)}{1.50\text{ m}} = \boxed{177\text{ N}}$$

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P10.32 (a) See ANS. FIG. P10.32 below for the force diagrams. For m_1, $\sum F_y = ma_y$ gives

$$+n - m_1 g = 0$$

$$n_1 = m_1 g$$

with $f_{k1} = \mu_k n_1$.

$\sum F_x = ma_x$ gives

$$-f_{k1} + T_1 = m_1 a \qquad \textbf{[1]}$$

For the pulley, $\sum \tau = I\alpha$ gives

$$-T_1 R + T_2 R = \frac{1}{2} MR^2 \left(\frac{a}{R} \right)$$

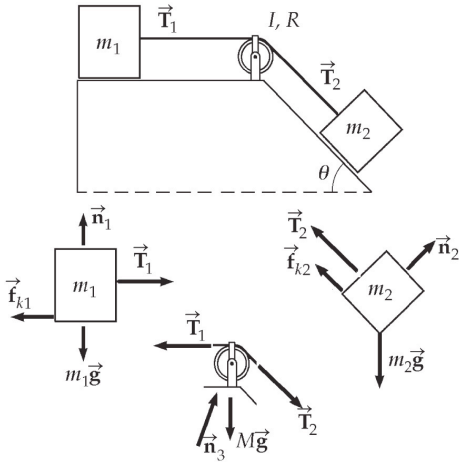
or $$-T_1 + T_2 = \frac{1}{2} MR \left(\frac{a}{R} \right) \rightarrow -T_1 + T_2 = \frac{1}{2} Ma \qquad \textbf{[2]}$$

For m_2,

$$+n_2 - m_2 g \cos\theta = 0 \rightarrow n_2 = m_2 g \cos\theta$$

$$f_{k2} = \mu_k n_2$$

$$-f_{k2} - T_2 + m_2 g \sin\theta = m_2 a \qquad \textbf{[3]}$$



ANS. FIG. P10.32

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(b) Add equations [1], [2], and [3] and substitute the expressions for f_{k1} and n_1, and $-f_{k2}$ and n_2:

$$-f_{k1} + T_1 + (-T_1 + T_2) - f_{k2} - T_2 + m_2 g \sin\theta = m_1 a + \frac{1}{2} Ma + m_2 a$$

$$-f_{k1} - f_{k2} + m_2 g \sin\theta = \left(m_1 + m_2 + \frac{1}{2} M \right) a$$

$$-\mu_k m_1 g - \mu_k m_2 g \cos\theta + m_2 g \sin\theta = \left(m_1 + m_2 + \frac{1}{2} M \right) a$$

$$a = \frac{m_2 (\sin\theta - \mu_k \cos\theta) - \mu_k m_1}{m_1 + m_2 + \frac{1}{2} M} g$$

$$a = \frac{(6.00 \text{ kg})(\sin 30.0^\circ - 0.360 \cos 30.0^\circ) - 0.360(2.00 \text{ kg})}{(2.00 \text{ kg}) + (6.00 \text{ kg}) + \frac{1}{2}(10.0 \text{ kg})} g$$

$$a = \boxed{0.309 \text{ m/s}^2}$$

(c) From equation [1]:

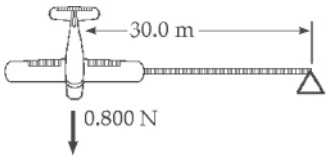
$$-f_{k1} + T_1 = m_1 a \rightarrow T_1 = 2.00 \text{ kg}(0.309 \text{ m/s}^2) + 7.06 \text{ N} = \boxed{7.67 \text{ N}}$$

From equation [2]:

$$-T_1 + T_2 = \frac{1}{2} Ma \rightarrow T_2 = 7.67 \text{ N} + 5.00 \text{ kg}(0.309 \text{ m/s}^2)$$

$$= \boxed{9.22 \text{ N}}$$

P10.33 We use the definition of torque and the relationship between angular and translational acceleration, with $m = 0.750$ kg and $F = 0.800$ N:



ANS. FIG. P10.33

(a) $\tau = rF = (30.0 \text{ m})(0.800 \text{ N}) = \boxed{24.0 \text{ N} \cdot \text{m}}$

(b) $$\alpha = \frac{\tau}{I} = \frac{rF}{mr^2} = \frac{24.0 \text{ N} \cdot \text{m}}{(0.750 \text{ kg})(30.0 \text{ m})^2}$$

$$= \boxed{0.035\,6 \text{ rad/s}^2}$$

(c) $a_t = \alpha r = (0.035\,6 \text{ rad/s}^2)(30.0 \text{ m}) = \boxed{1.07 \text{ m/s}^2}$

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P10.34 (a) The chosen tangential force produces constant torque and therefore constant angular acceleration. Since the disk starts from rest, we write

$$\theta_f - \theta_i = \omega_i t + \frac{1}{2}\alpha t^2$$

$$\theta_f - 0 = 0 + \frac{1}{2}\alpha t^2$$

$$\theta_f = \frac{1}{2}\alpha t^2$$

Solving for the angular acceleration gives

$$\alpha = \frac{2\theta_f}{t^2} = \frac{2(2.00 \text{ rev})\left(\frac{2\pi \text{ rad}}{1 \text{ rev}}\right)}{(10.0 \text{ s})^2} = 0.251 \text{ rad/s}^2$$

We then obtain the required combination of F and R from the rigid object under a net torque model:

$$\sum \tau = I\alpha: \qquad FR = \left(100 \text{ kg} \cdot \text{m}^2\right)(0.251 \text{ rad/s}^2) = 25.1 \text{ N} \cdot \text{m}$$

For $F = 25.1$ N, $R = 1.00$ m. For $F = 10.0$ N, $R = 2.51$ m.

(b) No. Infinitely many pairs of values that satisfy this requirement exist: for any $F \leq 50.0$ N, $R = 25.1 \text{ N} \cdot \text{m}/F$, as long as $R \leq 3.00$ m.

P10.35 (a) From the rigid object under a net torque model, $\sum \tau = I\alpha$ gives

$$I = \frac{\sum \tau}{\alpha} = \frac{\sum \tau}{\Delta\omega}\Delta t = \frac{36.0 \text{ N} \cdot \text{m}}{10.0 \text{ rad/s}}(6.00 \text{ s}) = \boxed{21.6 \text{ kg} \cdot \text{m}^2}$$

(b) For the portion of the motion during which the wheel slows down,

$$\left|\sum \tau\right| = |I\alpha| = \left|I\frac{\Delta\omega}{\Delta t}\right| = \left|\left(21.6 \text{ kg} \cdot \text{m}^2\right)\left(\frac{-10.0 \text{ rad/s}}{60.0 \text{ s}}\right)\right|$$

$$= \boxed{3.60 \text{ N} \cdot \text{m}}$$

(c) During the first portion of the motion,

$$\Delta\theta = \omega_{\text{avg}}\Delta t = \left(\frac{\omega_i + \omega_f}{2}\right)\Delta t = \left(\frac{0 + 10.0 \text{ rad/s}}{2}\right)(6.00 \text{ s})$$

$$= 30 \text{ rad}$$

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During the second portion,

$$\Delta\theta = \omega_{\text{avg}}\Delta t = \left(\frac{\omega_i + \omega_f}{2}\right)\Delta t = \left(\frac{10.0 \text{ rad/s} + 0}{2}\right)(60.0 \text{ s})$$
$$= 300 \text{ rad}$$

Therefore, the total angle is 330 rad or $\boxed{52.5 \text{ revolutions}}$.

P10.36 (a) Let T_1 represent the tension in the cord above m_1 and T_2 the tension in the cord above the lighter mass. The two blocks move with the same acceleration because the cord does not stretch, and the angular acceleration of the pulley is a/R. For the heavier mass we have

$$\sum F = m_1 a \rightarrow T_1 - m_1 g = m_1(-a) \quad \text{or} \quad -T_1 + m_1 g = m_1 a$$

For the lighter mass,

$$\sum F = m_2 a \rightarrow T_2 - m_2 g = m_2 a$$

We assume the pulley is a uniform disk: $I = (1/2)MR^2$

$$\sum \tau = I\alpha \rightarrow + T_1 R - T_2 R = \frac{1}{2} MR^2 (a/R)$$

or $$T_1 - T_2 = \frac{1}{2} Ma$$

Add up the three equations in a:

$$-T_1 + m_1 g + T_2 - m_2 g + T_1 - T_2 = m_1 a + m_2 a + \frac{1}{2} Ma$$

$$a = \frac{m_1 - m_2}{m_1 + m_2 + \frac{1}{2}M} g$$
$$= \frac{20.0 \text{ kg} - 12.5 \text{ kg}}{20.0 \text{ kg} + 12.5 \text{ kg} + \frac{1}{2}(5.00 \text{ kg})}(9.80 \text{ m/s}^2)$$
$$= 2.10 \text{ m/s}^2$$

Next, $x = 0 + 0 + \frac{1}{2}at^2 \rightarrow t = \sqrt{\frac{2x}{a}} = \sqrt{\frac{2(4.00 \text{ m})}{2.10 \text{ m/s}^2}} = \boxed{1.95 \text{ s}}$

(b) If the pulley were massless, the acceleration would be larger by a factor 35/32.5 and the time shorter by the square root of the factor 32.5/35. That is, the time would be reduced by 3.64%.

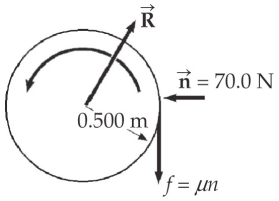
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P10.37 From the rigid object under a net torque model,

$$\sum \tau = I\alpha$$

$$-f_k R = \mu_k F R = \left(\frac{1}{2}MR^2\right)\frac{\Delta\omega}{\Delta t}$$

$$\mu_k = -\frac{MR\Delta\omega}{2F\Delta t}$$



ANS. FIG. P10.37

Substitute numerical values:

$$\mu_k = -\frac{(100\text{ kg})(0.500\text{ m})(-50.0\text{ rev/min})}{2(70.0\text{ N})(6.00\text{ s})}\left(\frac{2\pi\text{ rad}}{1\text{ rev}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)$$

$$= \boxed{0.312}$$

Section 10.6 Calculation of Moments of Inertia

P10.38 Model your body as a cylinder of mass 60.0 kg and a radius of 12.0 cm. Then its moment of inertia is

$$\frac{1}{2}MR^2 = \frac{1}{2}(60.0\text{ kg})(0.120\text{ m})^2 = 0.432\text{ kg}\cdot\text{m}^2$$

$$\sim \boxed{10^0\text{ kg}\cdot\text{m}^2 = 1\text{ kg}\cdot\text{m}^2}$$

P10.39 (a) Every particle in the door could be slid straight down into a high-density rod across its bottom, without changing the particle's distance from the rotation axis of the door. Thus, a rod 0.870 m long with mass 23.0 kg, pivoted about one end, has the same rotational inertia as the door:

$$I = \frac{1}{3}ML^2 = \frac{1}{3}(23.0\text{ kg})(0.870\text{ m})^2 = \boxed{5.80\text{ kg}\cdot\text{m}^2}$$

(b) The [height of the door is unnecessary] data.

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P10.40 (a) We take a coordinate system with mass M at the origin. The distance from the axis to the origin is also x. The moment of ineria about the axis is

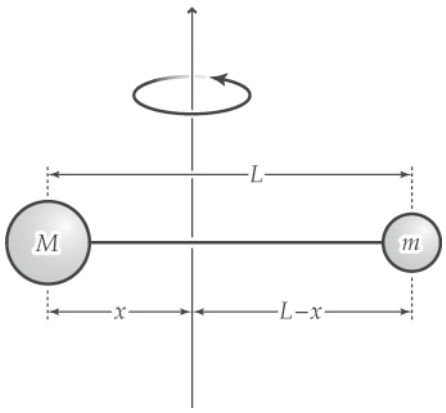
$$I = Mx^2 + m(L-x)^2$$

To find the extrema in the moment of inertia, we differentiate I with respect to x:

$$\frac{dI}{dx} = 2Mx - 2m(L-x) = 0$$

Solving for x then gives

$$x = \frac{mL}{M+m}$$



ANS. FIG. P10.40

Differentiating again gives $\frac{d^2I}{dx^2} = 2m + 2M$; therefore, I is at a minimum when the axis of rotation passes through $x = \frac{mL}{M+m}$, which is also the position of the center of mass of the system if we take mass M to lie at the origin of a coordinate system.

(b) The moment of inertia about an axis passing through x is

$$I_{CM} = M\left[\frac{mL}{M+m}\right]^2 + m\left[1 - \frac{m}{M+m}\right]^2 L^2 = \frac{Mm}{M+m}L^2$$

$$\rightarrow \boxed{I_{CM} = \mu L^2, \text{ where } \mu = \frac{Mm}{M+m}}$$

P10.41 Treat the tire as consisting of three hollow cylinders: two sidewalls and a tread region. The moment of inertia of a hollow cylinder, where $R_2 > R_1$, is $I = \frac{1}{2}M\left(R_1^2 + R_2^2\right)$, and the mass of a hollow cylinder of height (or thickness) t is $M = \rho\pi\left(R_2^2 - R_1^2\right)t$. Substituting the expression for mass M into the expression for I, we get

$$I = \frac{1}{2}\rho\pi\left(R_2^2 - R_1^2\right)t\left(R_1^2 + R_2^2\right) = \frac{1}{2}\rho\pi t\left(R_2^4 - R_1^4\right)$$

The two sidewalls have inner radius r_1 = 16.5 cm, outer radius r_2 = 30.5 cm, and height t_{side} = 0.635 cm. The tread region has inner radius r_2 = 30.5 cm, outer radius r_3 =33.0 cm, and height t_{tread} = 20.0 cm. The

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density of the rubber is 1.10×10^3 kg/m^3.

For the tire (two sidewalls: $R_1 = r_1$, $R_2 = r_2$; tread region: $R_1 = r_2$, $R_2 = r_3$)

$$I_{\text{total}} = 2\left[\frac{1}{2}\rho\pi t_{\text{side}}\left(R_2^4 - R_1^4\right)\right] + \frac{1}{2}\rho\pi t_{\text{tread}}\left(R_2^4 - R_1^4\right)$$

$$= 2\left[\frac{1}{2}\rho\pi t_{\text{side}}\left(r_2^4 - r_1^4\right)\right] + \frac{1}{2}\rho\pi t_{\text{tread}}\left(r_3^4 - r_2^4\right)$$

Substituting,

$$I_{\text{total}} = 2\left\{\frac{1}{2}\left(1.10\times10^3\ \text{kg/m}^3\right)\pi\left(6.35\times10^{-3}\ \text{m}\right)\right.$$

$$\left.\times\left[(0.305\ \text{m})^4 - (0.165\ \text{m})^4\right]\right\}$$

$$+\frac{1}{2}\left(1.10\times10^3\ \text{kg/m}^3\right)\pi(0.200\ \text{m})\left[(0.330\ \text{m})^4 - (0.305\ \text{m})^4\right]$$

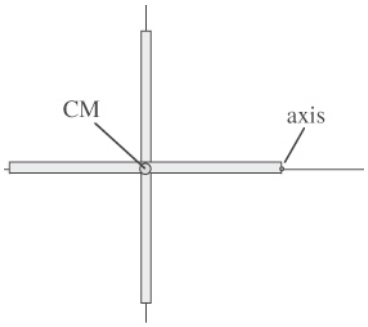
$$= 2\left(8.68\times10^{-2}\ \text{kg}\cdot\text{m}^2\right) + 1.11\ \text{kg}\cdot\text{m}^2 = \boxed{1.28\ \text{kg}\cdot\text{m}^2}$$

P10.42 We use x as a measure of the distance of each mass element dm in the rod from the y' axis:

$$\boxed{I_{y'} = \int_{\text{all mass}} r^2 dm = \int_0^L x^2 \frac{M}{L} dx = \frac{M}{L}\frac{x^3}{3}\bigg|_0^L = \frac{1}{3}ML^2}$$

P10.43 We assume the rods are thin, with radius much less than L. Note that the center of mass (CM) of the rod combination lies at the origin of the coordinate system. Because the axis of rotation is parallel to the y axis, we can first calculate the moment of inertia of the rods about the y axis, then use the parallel-axis theorem to find the moment about the axis of rotation.

The moment of the rod on the y axis about the y axis itself is essentially zero (axis through center, parallel to rod) because the rod is thin. The moments of the rods on the x and z axes are each $I = \frac{1}{12}mL^2$ (axis through center, perpendicular to rod) from the table in the chapter.



ANS. FIG. P10.43

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The total moment of the three rods about the y axis (and about the CM) is

$$I_{CM} = I_{\text{on } x \text{ axis}} + I_{\text{on } y \text{ axis}} + I_{\text{on } z \text{ axis}}$$
$$= \frac{1}{12}mL^2 + 0 + \frac{1}{12}mL^2 = \frac{1}{6}mL^2$$

For the moment of the rod-combination about the axis of rotation, the parallel-axis theorem gives

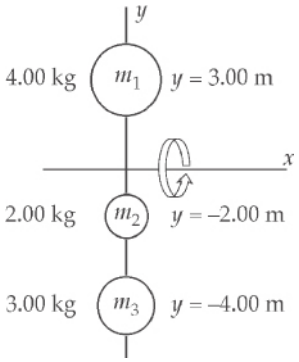
$$I = I_{CM} + 3m\left(\frac{L}{2}\right)^2 = \left[\frac{1}{6} + \frac{3}{4}\right]mL^2 = \left[\frac{2}{12} + \frac{9}{12}\right]mL^2 = \boxed{\frac{11}{12}mL^2}$$

Section 10.7 Rotational Kinetic Energy

P10.44 The masses and distances from the rotation axis for the three particles are:

$$m_1 = 4.00 \text{ kg},\ r_1 = |y_1| = 3.00 \text{ m}$$
$$m_2 = 2.00 \text{ kg},\ r_2 = |y_2| = 2.00 \text{ m}$$
$$m_3 = 3.00 \text{ kg},\ r_3 = |y_3| = 4.00 \text{ m}$$

and $\omega = 2.00 \text{ rad/s}$ about the x axis.



ANS. FIG. P10.44

(a) $I_x = m_1r_1^2 + m_2r_2^2 + m_3r_3^2$

$$I_x = (4.00 \text{ kg})(3.00 \text{ m})^2 + (2.00 \text{ kg})(2.00 \text{ m})^2 + (3.00 \text{ kg})(4.00 \text{ m})^2$$
$$= \boxed{92.0 \text{ kg} \cdot \text{m}^2}$$

(b) $K_R = \frac{1}{2}I_x\omega^2 = \frac{1}{2}(92.0 \text{ kg} \cdot \text{m}^2)(2.00 \text{ m})^2 = \boxed{184 \text{ J}}$

(c) $v_1 = r_1\omega = (3.00 \text{ m})(2.00 \text{ rad/s}) = \boxed{6.00 \text{ m/s}}$

$v_2 = r_2\omega = (2.00 \text{ m})(2.00 \text{ rad/s}) = \boxed{4.00 \text{ m/s}}$

$v_3 = r_3\omega = (4.00 \text{ m})(2.00 \text{ rad/s}) = \boxed{8.00 \text{ m/s}}$

(d) $K_1 = \frac{1}{2}m_1v_1^2 = \frac{1}{2}(4.00 \text{ kg})(6.00 \text{ m/s})^2 = 72.0 \text{ J}$

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$$K_2 = \frac{1}{2}m_2 v_2^2 = \frac{1}{2}(2.00\text{ kg})(4.00\text{ m/s})^2 = 16.0\text{ J}$$

$$K_3 = \frac{1}{2}m_3 v_3^2 = \frac{1}{2}(3.00\text{ kg})(8.00\text{ m/s})^2 = 96.0\text{ J}$$

$$K = K_1 + K_2 + K_3 = 72.0\text{ J} + 16.0\text{ J} + 96.0\text{ J} = \boxed{184\text{ J}} = \frac{1}{2}I_x\omega^2$$

(e) The kinetic energies computed in parts (b) and (d) are the same.

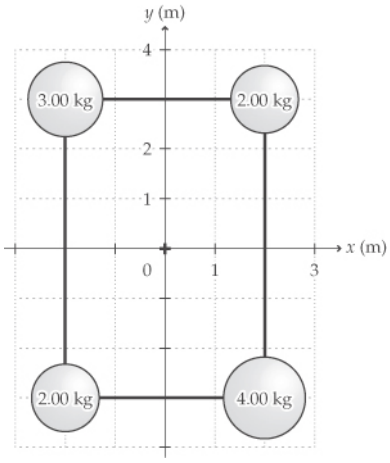
Rotational kinetic energy of an object rotating about a fixed axis can be viewed as the total translational kinetic energy of the particles moving in circular paths.

P10.45 (a) All four particles are at a distance r from the z axis, with

$$r^2 = (3.00\text{ m})^2 + (2.00\text{ m})^2$$
$$= 13.0\text{ m}^2$$

Thus the moment of inertia is

$$I_z = \sum m_i r_i^2$$
$$= (3.00\text{ kg})(13.0\text{ m}^2)$$
$$+ (2.00\text{ kg})(13.0\text{ m}^2)$$
$$+ (4.00\text{ kg})(13.0\text{ m}^2)$$
$$+ (2.00\text{ kg})(13.0\text{ m}^2)$$
$$= \boxed{143\text{ kg}\cdot\text{m}^2}$$



ANS. FIG. P10.45

(b) The rotational kinetic energy of the four-particle system is

$$K_R = \frac{1}{2}I\omega^2 = \frac{1}{2}(143\text{ kg}\cdot\text{m}^2)(6.00\text{ rad/s})^2 = \boxed{2.57\times10^3\text{ J}}$$

P10.46 The cam is a solid disk of radius R that has had a small disk of radius $R/2$ cut from it. To find the moment of inertia of the cam, we use the parallel-axis theorem to find the moment of inertia of the solid disk about an axis at distance $R/2$ from its CM, then subtract off the moment of inertia of the small disk of radius $R/2$ with axis through its center.

By the parallel-axis theorem, the moment of inertia of the solid disk about an axis $R/2$ from its CM is

$$I_{\text{disk}} = I_{\text{CM}} + M_{\text{disk}}\left(\frac{R}{2}\right)^2 = \frac{1}{2}M_{\text{disk}}R^2 + \frac{1}{4}M_{\text{disk}}R^2 = \frac{3}{4}M_{\text{disk}}R^2$$

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With half the radius, the cut-away small disk has one-quarter the face area and one-quarter the volume and one-quarter the mass M_{disk} of the original solid disk:

$$\frac{M_{\text{small disk}}}{M_{\text{disk}}} = \frac{(R/2)^2}{R^2} = \frac{1}{4}$$

The moment of inertia of the small disk of radius $R/2$ about an axis through its CM is

$$I_{\text{small disk}} = \frac{1}{2} M_{\text{small disk}} \left(\frac{R}{2}\right)^2 = \frac{1}{2}\left[\frac{1}{4} M_{\text{disk}}\right]\frac{R^2}{4} = \frac{1}{32} M_{\text{disk}} R^2$$

Subtracting the moment of the small disk from the solid disk, we find for the cam

$$I_{\text{cam}} = I_{\text{disk}} - I_{\text{small disk}} = \frac{3}{4} M_{\text{disk}} R^2 - \frac{1}{32} M_{\text{disk}} R^2$$

$$I_{\text{cam}} = M_{\text{disk}} R^2 \left[\frac{24}{32} - \frac{1}{32}\right] = \frac{23}{32} M_{\text{disk}} R^2$$

The mass of the cam is $M = M_{\text{disk}} - M_{\text{small disk}} = M_{\text{disk}} - \frac{1}{4} M_{\text{disk}} = \frac{3}{4} M_{\text{disk}}$, therefore

$$I_{\text{cam}} = \frac{23}{32} M_{\text{disk}} R^2 \left(\frac{M}{\frac{3}{4} M_{\text{disk}}}\right) = MR^2 \left(\frac{23}{32}\right)\left(\frac{4}{3}\right) = \frac{23}{24} MR^2$$

The moment of inertia of the cam-shaft is the sum of the moments of the cam and the shaft:

$$I_{\text{cam-shaft}} = I_{\text{cam}} + I_{\text{shaft}} = \frac{23}{24} MR^2 + \frac{1}{2} M \left(\frac{R}{2}\right)^2$$

$$= MR^2 \left[\frac{23}{24} + \frac{1}{8}\right] = MR^2 \left[\frac{23}{24} + \frac{3}{24}\right]$$

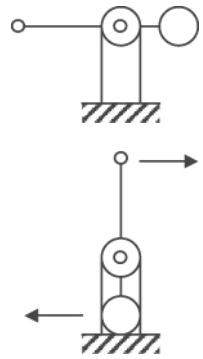
$$I_{\text{cam-shaft}} = \frac{26}{24} MR^2 = \frac{13}{12} MR^2$$

The kinetic energy of the cam-shaft combination rotating with angular speed ω is

$$K = \frac{1}{2} I_{\text{can-shaft}} \omega^2 = \frac{1}{2}\left(\frac{13}{12} MR^2\right)\omega^2 = \boxed{\frac{13}{24} MR^2 \omega^2}$$

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P10.47 (a) Identify the two objects and the Earth as an isolated system. The maximum speed of the lighter object will occur when the rod is in the vertical position so let's define the time interval as from when the system is released from rest to when the rod reaches a vertical orientation. So, for the isolated system,



ANS. FIG. P10.47

$$\Delta K + \Delta U = 0$$

$$\left[\left(\frac{1}{2}I_1\omega^2 + \frac{1}{2}I_2\omega^2\right) - 0\right] + \left[m_1 g y_1 + m_2 g y_2 - 0\right] = 0$$

$$\omega = \sqrt{\frac{-2g\left(m_1 y_1 + m_2 y_2\right)}{I_1 + I_2}} = \sqrt{\frac{-2g\left(m_1 y_1 + m_2 y_2\right)}{m_1 r_1^2 + m_2 r_2^2}}$$

$$= \sqrt{\frac{-2\left(9.80\ \text{m/s}^2\right)\left[\left(0.120\ \text{kg}\right)\left(2.86\ \text{m}\right) + \left(60.0\ \text{kg}\right)\left(-0.140\ \text{m}\right)\right]}{\left(0.120\ \text{kg}\right)\left(2.86\ \text{m}\right)^2 + \left(60.0\ \text{kg}\right)\left(0.140\ \text{m}\right)^2}}$$

$$= 8.55\ \text{rad/s}$$

Then, the tangential speed of the lighter object is,

$$v = r\omega = \left(2.86\ \text{m}\right)\left(8.55\ \text{rad/s}\right) = \boxed{24.5\ \text{m/s}}$$

(b) No. The overall acceleration is not constant. It has to move either in a straight line or parabolic path to have a chance of being under constant acceleration. The circular path presented here rules out that possibility.

(c) No. It does not move with constant tangential acceleration, since the angular acceleration is not constant. See explanation in part (d).

(d) No. The lever arm of the gravitational force acting on the 60-kg mass changes during the motion. As a result, the torque changes, and so does the angular acceleration.

(e) No. The angular velocity changes, therefore the angular momentum of the trebuchet changes.

(f) Yes. The mechanical energy stays constant because the system is isolated—that is how we solved the problem in (a).

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Section 10.8 Energy Considerations in Rotational Motion

P10.48 From the rigid object under a net torque model,

$$\sum \tau = I\alpha \quad \rightarrow \quad \alpha = \frac{\sum \tau}{I} = \frac{FR}{\frac{1}{2}MR^2} = \frac{2F}{MR}$$

From the definition of rotational kinetic energy and the rigid object under constant angular acceleration model,

$$K = \frac{1}{2}I\omega^2 = \frac{1}{2}I\left(\omega_i + \alpha t\right)^2 = \frac{1}{2}I\alpha^2 t^2 = \frac{1}{2}\left(\frac{1}{2}MR^2\right)\left(\frac{2F}{MR}\right)^2 t^2$$

$$= \frac{F^2 t^2}{M}$$

Substituting,

$$K = \frac{(50.0\ \text{N})^2 (3.00\ \text{s})^2}{800\ \text{N}/9.80\ \text{m/s}^2} = \boxed{276\ \text{J}}$$

P10.49 The moment of inertia of a thin rod about an axis through one end is $I = \frac{1}{3}ML^2$. The total rotational kinetic energy is given as

$$K_R = \frac{1}{2}I_h\omega_h^2 + \frac{1}{2}I_m\omega_m^2$$

with $$I_h = \frac{m_h L_h^2}{3} = \frac{60.0\ \text{kg}(2.70\ \text{m})^2}{3} = 146\ \text{kg}\cdot\text{m}^2$$

and $$I_m = \frac{m_m L_m^2}{3} = \frac{100\ \text{kg}(4.50\ \text{m})^2}{3} = 675\ \text{kg}\cdot\text{m}^2$$

In addition,

$$\omega_h = \frac{2\pi\ \text{rad}}{12\ \text{h}}\left(\frac{1\ \text{h}}{3\,600\ \text{s}}\right) = 1.45\times10^{-4}\ \text{rad/s}$$

while $$\omega_m = \frac{2\pi\ \text{rad}}{1\ \text{h}}\left(\frac{1\ \text{h}}{3\,600\ \text{s}}\right) = 1.75\times10^{-3}\ \text{rad/s}$$

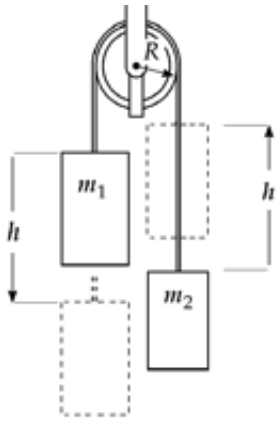
Therefore,

$$K_R = \frac{1}{2}\left(146\ \text{kg}\cdot\text{m}^2\right)\left(1.45\times10^{-4}\ \text{rad/s}\right)^2 + \frac{1}{2}\left(675\ \text{kg}\cdot\text{m}^2\right)\left(1.75\times10^{-3}\ \text{rad/s}\right)^2$$

$$= \boxed{1.04\times10^{-3}\ \text{J}}$$

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***P10.50** Take the two objects, pulley, and Earth as the system. If we neglect friction in the system, then mechanical energy is conserved and we can state that the increase in kinetic energy of the system equals the decrease in potential energy. Since $K_i = 0$ (the system is initially at rest), we have



ANS. FIG. P10.50

$$\Delta K = K_f - K_i$$

$$= \frac{1}{2}m_1 v^2 + \frac{1}{2}m_2 v^2 + \frac{1}{2}I\omega^2$$

where m_1 and m_2 have a common speed. But $v = R\omega$ so that $\Delta K = \frac{1}{2}\left(m_1 + m_2 + \frac{I}{R^2}\right)v^2$.

From ANS. FIG. P10.50, we see that the system loses potential energy because of the motion of m_1 and gains potential energy because of the motion of m_2. Applying the law of conservation of energy, $\Delta K + \Delta U = 0$, gives

$$\frac{1}{2}\left(m_1 + m_2 + \frac{I}{R^2}\right)v^2 + m_2 gh - m_1 gh = 0$$

$$\boxed{v = \sqrt{\frac{2(m_1 - m_2)gh}{m_1 + m_2 + \frac{I}{R^2}}}}$$

Since $v = R\omega$, the angular speed of the pulley at this instant is given by

$$\boxed{\omega = \frac{v}{R} = \sqrt{\frac{2(m_1 - m_2)gh}{m_1 R^2 + m_2 R^2 + I}}}$$

P10.51 For the nonisolated system of the top,

$$W = \Delta K \quad \rightarrow \quad F\Delta x = \left(\frac{1}{2}I\omega^2 - 0\right)$$

$$\rightarrow \quad \omega = \sqrt{\frac{2F\Delta x}{I}} = \sqrt{\frac{2(5.57\ \text{N})(0.800\ \text{m})}{4 \times 10^{-4}\ \text{kg} \cdot \text{m}^2}} = \boxed{149\ \text{rad/s}}$$

P10.52 The power output of the bus is $P = \frac{E}{\Delta t}$, where

$$E = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{1}{2}MR^2\omega^2\right) = \frac{1}{4}MR^2\omega^2$$

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is the stored energy and $\Delta t = \frac{d}{v}$ is the time it can roll. Then $\frac{1}{4}MR^2\omega^2 = P\Delta t = \frac{Pd}{v}$. The maximum range of the bus is then $d = \frac{MR^2\omega^2 v}{4P}$.

For average $P = (25.0 \text{ hp})\left(\frac{746 \text{ W}}{1 \text{ hp}}\right) = 18\,650 \text{ W}$ and average v =35.0 km/h = 9.72 m/s, the maximum range is

$$d = \frac{MR^2\omega^2 v}{4P}$$
$$= \frac{(1\,200 \text{ kg})(0.500 \text{ m})^2(3\,000 \cdot 2\pi / 60 \text{ s})^2(9.72 \text{ m/s})}{4(18\,650 \text{ W})}$$
$$= 3.86 \text{ km}$$

The situation is impossible because the range is only 3.86 km, not city-wide.

P10.53 (a) Apply $\Delta K + \Delta U + \Delta E_{\text{int}} = 0$, where $\Delta E_{\text{int}} = f_k d$, $\mu = 0.250$, and $f_k = \mu n_2 = \mu m_2 g$. Both translational and rotational kinetic energy are present in the system. $v_i = 0.820$ m/s. Find v. The angular speed of the pulley is $\omega_i = v_i / R_2$, and $\omega = v/R_2$. Mass m_1 drops by $h = d$ when mass m_2 moves distance $d = 0.700$ m.

$I = \frac{1}{2}M(R_1^2 + R_2^2)$, where $R_1 = 0.020\,0$ m, $R_2 = 0.030\,0$ m, and $M = 0.350$ kg.

$$(K_f - K_i) + (U_f - U_f) + \Delta E_{\text{int}} = 0$$
$$\left(\frac{1}{2}m_2v^2 - \frac{1}{2}m_2v_i^2\right) + \left(\frac{1}{2}m_1v^2 - \frac{1}{2}m_1v_i^2\right)$$
$$+ \left(\frac{1}{2}I\omega^2 - \frac{1}{2}I\omega_i^2\right) + (m_1gy - m_1gy_i) + f_k d = 0$$

$$\frac{1}{2}(m_1 + m_2)(v^2 - v_i^2) + \frac{1}{2}I\left[\left(\frac{v}{R_2}\right)^2 - \left(\frac{v_i}{R_2}\right)^2\right]$$
$$+ m_1g(y - y_i) + \mu m_2 g d = 0$$

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$$\frac{1}{2}(m_1+m_2)\left(v^2-v_i^2\right) + \frac{1}{2}\left[\frac{1}{2}M\left(R_1^2+R_2^2\right)\right]\left(\frac{1}{R_2}\right)^2\left[v^2-v_i^2\right] + m_1 g(-d)+\mu m_2 g d = 0$$

$$\frac{1}{2}(m_1+m_2)\left(v^2-v_i^2\right) + \frac{1}{2}\left[\frac{1}{2}M\left(1+\frac{R_1^2}{R_2^2}\right)\right]\left[v^2-v_i^2\right] = gd\left(m_1-\mu m_2\right)$$

$$\frac{1}{2}\left[(m_1+m_2)+\frac{1}{2}M\left(1+\frac{R_1^2}{R_2^2}\right)\right]\left(v^2-v_i^2\right) = gd\left(m_1-\mu m_2\right)$$

$$v = \left\{ v_i^{\,2} + \frac{4gd\left(m_1 - \mu_k m_2\right)}{2\left(m_1 + m_2\right) + M\left(1 + \frac{R_1^{\,2}}{R_2^{\,2}}\right)} \right\}^{1/2}$$

Suppressing units,

$$v = \left\{ (0.820)^2 + \frac{4(9.80)(0.700)\left[0.420 - (0.250)(0.850)\right]}{\left[2(0.420+0.850)+0.350\left(1+\frac{(0.020\ 0)^2}{(0.030\ 0)^2}\right)\right]} \right\}^{1/2}$$

$$= \boxed{1.59 \text{ m/s}}$$

(b) $\omega = \dfrac{v}{r} = \dfrac{1.59 \text{ m/s}}{0.030\ 0 \text{ m}} = \boxed{53.1 \text{ rad/s}}$

P10.54 (a) For the isolated rod-ball-Earth system,

$$\Delta K + \Delta U = 0 \rightarrow \left(K_f - 0\right) + \left(0 - U_i\right) = 0 \rightarrow K_f = U_i$$

$$K_f = m_{\text{rod}} g y_{\text{CM, rod}} + m_{\text{ball}} g y_{\text{CM, ball}}$$

$$= \left(m_{\text{rod}} y_{\text{CM, rod}} + m_{\text{ball}} y_{\text{CM, ball}}\right) g$$

$$= \left[(1.20 \text{ kg})(0.120 \text{ m}) + (2.00 \text{ kg})(0.280 \text{ m})\right]\left(9.80 \text{ m/s}^2\right)$$

$$= \boxed{6.90 \text{ J}}$$

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(b) We assume the rod is thin. For the compound object

$$I = \frac{1}{3}M_{\text{rod}}L^2 + \left[\frac{2}{5}m_{\text{ball}}R^2 + M_{\text{ball}}D^2\right]$$

$$= \frac{1}{3}(1.20\text{ kg})(0.240\text{ m})^2 + \frac{2}{5}(2.00\text{ kg})\left(4.00\times10^{-2}\text{ m}\right)^2 + (2.00\text{ kg})(0.280\text{ m})^2$$

$$I = 0.181\text{ kg}\cdot\text{m}^2$$

$$K_f = \frac{1}{2}\omega^2 \quad\rightarrow\quad \omega = \sqrt{\frac{2K_f}{I}} = \sqrt{\frac{2(6.90\text{ J})}{0.181\text{ kg}\cdot\text{m}^2}} = \boxed{8.73\text{ rad/s}}$$

(c) $v = r\omega = (0.280\text{ m})(8.73\text{ rad/s}) = \boxed{2.44\text{ m/s}}$

(d) $v_f^2 = v_i^2 + 2a\left(y_f - y_i\right)$

$$v_f = \sqrt{0 + 2\left(9.80\text{ m/s}^2\right)(0.280\text{ m})} = 2.34\text{ m/s}$$

The speed it attains in swinging is greater by

$$\frac{2.44}{2.34} = \boxed{1.043\,2\text{ times}}$$

P10.55 The gravitational force exerted on the reel is

$$mg = (5.10\text{ kg})\left(9.80\text{ m/s}^2\right) = 50.0\text{ N down}$$

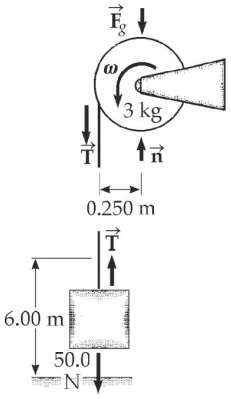
We use $\sum\tau = I\alpha$ to find T and a.

First find I for the reel, which we know is a uniform disk.

$$I = \frac{1}{2}MR^2 = \frac{1}{2}(3.00\text{ kg})(0.250\text{ m})^2$$

$$= 0.093\,8\text{ kg}\cdot\text{m}^2$$

The forces on the reel are shown in ANS. FIG. P10.55, including a normal force exerted by its axle. From the diagram, we can see that the tension is the only force that produces a torque causing the reel to rotate.



ANS. FIG. P10.55

$\sum\tau = I\alpha$ becomes

$$n(0) + F_{gp}(0) + T(0.250\text{ m}) = \left(0.093\,8\text{ kg}\cdot\text{m}^2\right)(a/0.250\text{ m}) \qquad \textbf{[1]}$$

where we have applied $a_t = r\alpha$ to the point of contact between string

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and reel. For the object that moves down,

$$\sum F_y = ma_y \qquad \text{becomes} \qquad 50.0\ \text{N} - T = (5.10\ \text{kg})a \qquad \textbf{[2]}$$

Note that we have defined downwards to be positive, so that positive linear acceleration of the object corresponds to positive angular acceleration of the reel. We now have our two equations in the unknowns T and a for the two connected objects. Substituting T from equation [2] into equation [1], we have

$$\left[50.0\ \text{N} - (5.10\ \text{kg})a\right](0.250\ \text{m}) = (0.093\ 8\ \text{kg}\cdot\text{m}^2)\left(\frac{a}{0.250\ \text{m}}\right)$$

(b) Solving for a from above gives

$$50.0\ \text{N} - (5.10\ \text{kg})a = (1.50\ \text{kg})a$$

$$a = \frac{50.0\ \text{N}}{6.60\ \text{kg}} = \boxed{7.57\ \text{m/s}^2}$$

Because we eliminated T in solving the simultaneous equations, the answer for a, required for part (b), emerged first. No matter—we can now substitute back to get the answer to part (a).

(a) $T = 50.0\ \text{N} - 5.10\ \text{kg}\ (7.57\ \text{m/s}^2) = \boxed{11.4\ \text{N}}$

(c) For the motion of the hanging weight,

$$v_f^2 = v_i^2 + 2a\left(y_f - y_i\right) = 0^2 + 2\left(7.57\ \text{m/s}^2\right)(6.00\ \text{m})$$

$$v_f = 9.53\ \text{m/s (down)}$$

(d) The isolated-system energy model can take account of multiple objects more easily than Newton's second law. Like your bratty cousins, the equation for conservation of energy grows between visits. Now it reads for the counterweight-reel-Earth system:

$$(K_1 + K_2 + U_g)_i = (K_1 + K_2 + U_g)_f$$

where K_1 is the translational kinetic energy of the falling object and K_2 is the rotational kinetic energy of the reel.

$$0 + 0 + m_1 g y_{1i} = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} I_2 \omega_{2f}^2 + 0$$

Now note that $\omega = v/r$ as the string unwinds from the reel.

$$mgy_i = \frac{1}{2} mv^2 + \frac{1}{2} I\omega^2$$

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$$2mgy_i = mv^2 + I\left(\frac{v^2}{R^2}\right) = v^2\left(m + \frac{I}{R^2}\right)$$

$$v = \sqrt{\frac{2mgy_i}{m + (I/R^2)}} = \sqrt{\frac{2(5.10\text{ kg})(9.80\text{ m/s}^2)(6.00\text{ m})}{5.10\text{ kg} + \dfrac{0.093\ 8\text{ kg}\cdot\text{m}^2}{(0.250\text{ m})^2}}}$$

$$= \boxed{9.53\text{ m/s}}$$

P10.56 Each point on the cord moves at a linear speed of $v = \omega r$, where r is the radius of the spool. The energy conservation equation for the counterweight-turntable-Earth system is:

$$(K_1 + K_2 + U_g)_i + W_{\text{other}} = (K_1 + K_2 + U_g)_f$$

Specializing, we have

$$0 + 0 + mgh + 0 = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 + 0$$

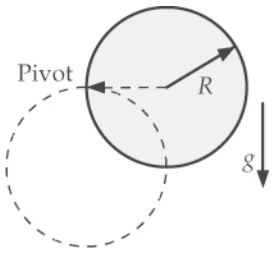
$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\frac{v^2}{r^2}$$

$$2mgh - mv^2 = I\frac{v^2}{r^2}$$

and finally,

$$I = \boxed{mr^2\left(\frac{2gh}{v^2} - 1\right)}$$

P10.57 To identify the change in gravitational energy, think of the height through which the center of mass falls. From the parallel-axis theorem, the moment of inertia of the disk about the pivot point on the circumference is



ANS. FIG. P10.57

$$I = I_{\text{CM}} + MD^2 = \frac{1}{2}MR^2 + MR^2$$

$$= \frac{3}{2}MR^2$$

The pivot point is fixed, so the kinetic energy is entirely rotational around the pivot. The equation for the isolated system (energy) model

$$(K + U)_i = (K + U)_f$$

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for the disk-Earth system becomes

$$0 + MgR = \frac{1}{2}\left(\frac{3}{2}MR^2\right)\omega^2 + 0$$

Solving for ω, $\omega = \sqrt{\frac{4g}{3R}}$

(a) At the center of mass, $v = R\omega = \boxed{2\sqrt{\frac{Rg}{3}}}$

(b) At the lowest point on the rim, $v = 2R\omega = \boxed{4\sqrt{\frac{Rg}{3}}}$

(c) For a hoop,

$$I_{CM} = MR^2 \quad \text{and} \quad I_{min} = 2MR^2$$

By conservation of energy for the hoop-Earth system, then

$$MgR = \frac{1}{2}\left(2MR^2\right)\omega^2 + 0$$

so $\omega = \sqrt{\frac{g}{R}}$

and the center of mass moves at $v_{CM} = R\omega = \boxed{\sqrt{gR}}$, slower than the disk.

P10.58 (a) The moment of inertia of the cord on the spool is

$$\frac{1}{2}M\left(R_1^2 + R_2^2\right) = \frac{1}{2}(0.100\ \text{kg})\left[(0.015\ 0\ \text{m})^2 + (0.090\ 0\ \text{m})^2\right]$$
$$= 4.16 \times 10^{-4}\ \text{kg} \cdot \text{m}^2$$

The protruding strand has mass

$$\left(1.00 \times 10^{-2}\ \text{kg/m}\right)(0.160\ \text{m}) = 1.60 \times 10^{-3}\ \text{kg}$$

and moment of inertia

$$I = I_{CM} + Md^2 = \frac{1}{12}ML^2 + Md^2$$
$$= \left(1.60 \times 10^{-3}\ \text{kg}\right)\left[\frac{1}{12}(0.160\ \text{m})^2 + (0.090\ 0\ \text{m} + 0.080\ 0\ \text{m})^2\right]$$
$$= 4.97 \times 10^{-5}\ \text{kg} \cdot \text{m}^2$$

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For the whole cord, $I = 4.66 \times 10^{-4}\ \text{kg} \cdot \text{m}^2$. In speeding up, the average power is

$$P = \frac{E}{\Delta t} = \frac{\frac{1}{2} I\omega^2}{\Delta t} = \left[\frac{4.66 \times 10^{-4}\ \text{kg} \cdot \text{m}^2}{2(0.215\ \text{s})} \right] \left(\frac{2\,500 \cdot 2\pi}{60\ \text{s}} \right)^2 = \boxed{74.3\ \text{W}}$$

(b) $$P = \tau\omega = (7.65\ \text{N})(0.160\ \text{m} + 0.090\,0\ \text{m}) \left(\frac{2\,000 \cdot 2\pi}{60\ \text{s}} \right) = \boxed{401\ \text{W}}$$

Section 10.9 Rolling Motion of a Rigid Object

P10.59 (a) The kinetic energy of translation is

$$K_{\text{trans}} = \frac{1}{2} mv^2 = \frac{1}{2}(10.0\ \text{kg})(10.0\ \text{m/s})^2 = \boxed{500\ \text{J}}$$

(b) Call the radius of the cylinder *r*. An observer at the center sees the rough surface and the circumference of the cylinder moving at 10.0 m/s, so the angular speed of the cylinder is

$$\omega = \frac{v_{\text{CM}}}{r} = \frac{10.0\ \text{m/s}}{r}$$

The moment of inertia about an axis through the center of mass is $I_{\text{CM}} = \frac{1}{2} mr^2$, so

$$K_{\text{rot}} = \frac{1}{2} I\omega^2 = \frac{1}{2}\left(\frac{1}{2} mr^2 \right)\left(\frac{v}{r} \right)^2 = \frac{1}{4}(10.0\ \text{kg})(10.0\ \text{m/s})^2$$
$$= \boxed{250\ \text{J}}$$

(c) We can now add up the total energy:

$$K_{\text{total}} = K_{\text{trans}} + K_{\text{rot}} = \boxed{750\ \text{J}}$$

P10.60 Conservation of energy for the sphere rolling without slipping is

$$U_i = K_{\text{translation},\,f} + K_{\text{rotation},\,f}$$

$$mgh = \frac{1}{2} mv^2 + \frac{1}{2}\left(\frac{2}{5} mR^2 \right)\left(\frac{v}{R} \right)^2 = \frac{7}{10} mv^2$$

which gives $$\boxed{v_f = \sqrt{\frac{10}{7} gh}}$$

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Conservation of energy for the sphere sliding without friction, with $\omega = 0$, is

$$mgh = \frac{1}{2}mv^2$$

which gives $\boxed{v_f = \sqrt{2gh}}$

The time intervals required for the trips follow from $x = 0 + v_{\text{avg}}t$:

$$\frac{h}{\sin\theta} = \left(\frac{0 + v_f}{2}\right)t \quad \rightarrow \quad t = \frac{2h}{v_f \sin\theta}$$

For rolling we have $t = \left(\frac{2h}{\sin\theta}\right)\sqrt{\frac{10}{7}gh}$

and for sliding, $t = \left(\frac{2h}{\sin\theta}\right)\sqrt{\frac{1}{2}gh}$

The time to roll is longer by a factor of $(0.7/0.5)^{1/2} = 1.18$.

P10.61 (a) We can consider the weight force acting at the center of mass (gravity) to exert a torque about the point of contact (the axis, in this case) between the disk and the incline. Then, from the particle under a net torque model, we have

$$\tau = I\alpha \quad \text{and} \quad a = R\alpha$$

$$mgR\sin\theta = \left(I_{\text{CM}} + mR^2\right)\alpha$$

$$a = \frac{mgR^2\sin\theta}{I_{\text{CM}} + mR^2}$$

$$a_{\text{disk}} = \frac{mgR^2\sin\theta}{\frac{3}{2}mR^2} = \boxed{\frac{2}{3}g\sin\theta}$$

(b) By the same method,

$$a = \frac{mgR^2\sin\theta}{I_{\text{CM}} + mR^2}$$

$\boxed{a_{\text{hoop}} = \frac{mgR^2\sin\theta}{2mR^2} = \frac{1}{2}g\sin\theta}$. The acceleration of the hoop is smaller than that of the disk.

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(c) Torque about the CM is caused by friction because the lever arm of the weight force is zero:

$$\tau = fR = I\alpha$$

$$f = \mu n = \mu mg\cos\theta$$

$$\mu = \frac{f}{mg\cos\theta} = \frac{I\alpha/R}{mg\cos\theta} = \frac{\left(\frac{2}{3}g\sin\theta\right)\left(\frac{1}{2}mR^2\right)}{R^2 mg\cos\theta} = \boxed{\frac{1}{3}\tan\theta}$$

P10.62 (a) Both systems of cube-Earth and cylinder-Earth are isolated; therefore, mechanical energy is conserved in both. The cylinder has extra kinetic energy, in the form of rotational kinetic energy, that is available to be transformed into potential energy, so it travels farther up the incline.

(b) The system of cube-Earth is isolated, so mechanical energy is conserved:

$$K_i = U_f \quad \rightarrow \quad \frac{1}{2}mv^2 = mgd\sin\theta \quad \rightarrow \quad d = \frac{v^2}{2g\sin\theta}$$

Static friction does no work on the cylinder because it acts at the point of contact and not through a distance; therefore, mechanical energy is conserved in the cylinder-Earth system:

$$K_{\text{translation, } i} + K_{\text{rotation, } i} = U_f \rightarrow \frac{1}{2}mv^2 + \frac{1}{2}\left[\frac{1}{2}mr^2\right]\left(\frac{v}{r}\right)^2 = mgd\sin\theta$$

which gives $d = \dfrac{3v^2}{4g\sin\theta}$.

The difference in distance is

$$\frac{3v^2}{4g\sin\theta} - \frac{v^2}{2g\sin\theta} = \boxed{\frac{v^2}{4g\sin\theta}}$$

or, the cylinder travels 50% farther.

(c) The cylinder does not lose mechanical energy because static friction does no work on it. Its rotation means that it has 50% more kinetic energy than the cube at the start, and so it travels 50% farther up the incline.

P10.63 (a) The disk reaches the bottom first because the ratio of its moment of inertia to its mass is smaller than for the hoop; this result is independent of the radius.

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(b) Both systems of disk-Earth and hoop-Earth are isolated because static friction does no work because it acts at the point of contact and not through a distance. Mechanical energy is conserved in both systems:

$$K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}\left[m + \frac{I}{R^2}\right]v^2$$

where $\omega = \dfrac{v}{R}$ since no slipping occurs.

Also, $U_i = mgh, U_f = 0,$ and $v_i = 0$

Therefore, $\dfrac{1}{2}\left[m + \dfrac{I}{R^2}\right]v^2 = mgh$

Thus, $v^2 = \dfrac{2gh}{\left[1 + \left(I/mR^2\right)\right]}$

For a disk, $I = \dfrac{1}{2}mR^2$, so $v^2 = \dfrac{2gh}{1+\frac{1}{2}}$ or $\boxed{v_{\text{disk}} = \sqrt{\dfrac{4gh}{3}}}$

For a hoop, $I = mR^2$ so $v^2 = \dfrac{2gh}{2}$ or $\boxed{v_{\text{hoop}} = \sqrt{gh}}$

Since $v_{\text{disk}} > v_{\text{hoop}}$, $\boxed{\text{the disk}}$ reaches the bottom first.

P10.64 (a) Energy conservation for the system of the ball and the Earth between the horizontal section and top of loop:

$$\frac{1}{2}mv_2^2 + \frac{1}{2}I\omega_2^2 + mgy_2 = \frac{1}{2}mv_1^2 + \frac{1}{2}I\omega_1^2$$

$$\frac{1}{2}mv_2^2 + \frac{1}{2}\left(\frac{2}{3}mr^2\right)\left(\frac{v_2}{r}\right)^2 + mgy_2 = \frac{1}{2}mv_1^2 + \frac{1}{2}\left(\frac{2}{3}mr^2\right)\left(\frac{v_1}{r}\right)^2$$

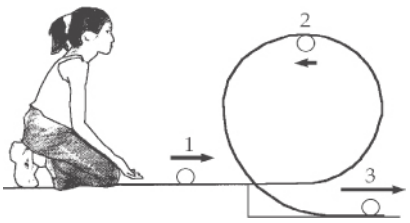
$$\frac{5}{6}v_2^2 + gy_2 = \frac{5}{6}v_1^2$$

$$v_2 = \sqrt{v_1^2 - \frac{6}{5}gy_2}$$

$$= \sqrt{\left(4.03\ \text{m/s}\right)^2 - \frac{6}{5}\left(9.80\ \text{m/s}^2\right)\left(0.900\ \text{m}\right)}$$

$$= \boxed{2.38\ \text{m/s}}$$

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ANS. FIG. P10.64

(b) The centripetal acceleration at the top is

$$\frac{v_2^2}{r} = \frac{(2.38\ \text{m/s})^2}{0.450\ \text{m}} = 12.6\ \text{m/s}^2 > g$$

Thus, the ball must be in contact with the track, with the track pushing downward on it.

(c) $$\frac{1}{2}mv_3^2 + \frac{1}{2}\left(\frac{2}{3}mr^2\right)\left(\frac{v_3}{r}\right)^2 + mgy_3 = \frac{1}{2}mv_1^2 + \frac{1}{2}\left(\frac{2}{3}mr^2\right)\left(\frac{v_1}{r}\right)^2$$

$$v_3 = \sqrt{v_1^2 - \frac{6}{5}gy_3}$$

$$= \sqrt{(4.03\ \text{m/s})^2 - \frac{6}{5}(9.80\ \text{m/s}^2)(-0.200\ \text{m})}$$

$$= \boxed{4.31\ \text{m/s}}$$

(d) $$\frac{1}{2}mv_2^2 + mgy_2 = \frac{1}{2}mv_1^2$$

$$v_2 = \sqrt{v_1^2 - 2gy_2} = \sqrt{(4.03\ \text{m/s})^2 - 2(9.80\ \text{m/s}^2)(0.900\ \text{m})}$$

$$= \boxed{\sqrt{-1.40\ \text{m}^2/\text{s}^2}}\ !$$

(e) This result is imaginary. In the case where the ball does not roll, the ball starts with less kinetic energy than in part (a) and never makes it to the top of the loop.

P10.65 (a) For the isolated can-Earth system,

$$\Delta K + \Delta U = 0 \ \rightarrow\ \left(\frac{1}{2}mv_{\text{CM}}^2 + \frac{1}{2}I\omega^2 - 0\right) + (0 - mgh) = 0$$

which gives

$$I = \frac{2mgh - mv_{\text{CM}}^2}{\omega^2} = (2mgh - mv_{\text{CM}}^2)\left(\frac{r^2}{v_{\text{CM}}^2}\right) = mr^2\left(\frac{2gh}{v_{\text{CM}}^2} - 1\right)$$

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From the particle under constant acceleration model,

$$v_{\text{CM, avg}} = \frac{0 + v_{\text{CM}}}{2} \rightarrow v_{\text{CM}} = 2v_{\text{CM, avg}} = \frac{2d}{\Delta t}$$

Therefore, the moment of inertia is

$$I = mr^2\left(\frac{2gh(\Delta t)^2}{4d^2} - 1\right) = mr^2\left(\frac{2g(d\sin\theta)(\Delta t)^2}{4d^2} - 1\right)$$

$$= mr^2\left(\frac{g(\sin\theta)(\Delta t)^2}{2d} - 1\right)$$

Substitute numerical values:

$$I = (0.215\ \text{kg})(0.031\ 9\ \text{m})^2 \times\left(\frac{(9.80\ \text{m/s}^2)(\sin 25.0°)(1.50\ \text{s})^2}{2(3.00\ \text{m})} - 1\right)$$

$$= \boxed{1.21 \times 10^{-4}\ \text{kg} \cdot \text{m}^2}$$

(b) The **height of the can** is unnecessary data.

(c) **The mass is not uniformly distributed; the density of the metal can is larger than that of the soup.**

Additional Problems

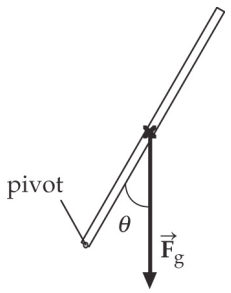
P10.66 When the rod is at angle θ from the vertical, the vertical weight force mg is at the same angle from the vertical so that its torque about the pivot is $mg\frac{\ell}{2}\sin\theta$. From the particle under a net torque model,

$$\sum \tau = I\alpha$$

$$mg\frac{\ell}{2}\sin\theta = \frac{1}{3}m\ell^2\alpha$$

$$\alpha = \frac{3}{2}\frac{g}{\ell}\sin\theta \rightarrow a_t = \left(\frac{3}{2}\frac{g}{\ell}\sin\theta\right)r$$

For $\left(\frac{3}{2}\frac{g}{\ell}\sin\theta\right)r > g\sin\theta \rightarrow r > \frac{2}{3}\ell$



ANS. FIG. P10.66

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$\therefore$ About $\boxed{\frac{1}{3} \text{ the length of the chimney}}$ will have a tangential acceleration greater than $g\sin\theta$.

P10.67 (a) The spool starts from rest, with zero rotational kinetic energy, and accelerates to 8.00 rad/s. The work done to accomplish this is given by the work-kinetic energy theorem:

$$W = \Delta K = \frac{1}{2}I\omega_f^2 - \frac{1}{2}I\omega_i^2 = \frac{1}{2}I\left(\omega_f^2 - \omega_i^2\right), \quad \text{where} \quad I = \frac{1}{2}mR^2$$

$$= \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)(1.00\text{ kg})(0.500\text{ m})^2\left[(8.00\text{ rad/s})^2 - 0\right] = \boxed{4.00\text{ J}}$$

(b) The time interval can be found from

$$\omega_f = \omega_i + \alpha t, \quad \text{where} \quad \alpha = \frac{a}{r} = \frac{2.50\text{ m/s}^2}{0.500\text{ m}} = 5.00\text{ rad/s}^2$$

Therefore,

$$t = \frac{\omega_f - \omega_i}{\alpha} = \frac{8.00\text{ rad/s} - 0}{5.00\text{ rad/s}^2} = \boxed{1.60\text{ s}}$$

(c) The spool turns through angular displacement

$$\theta_f = \theta_i + \omega_i t + \frac{1}{2}\alpha t^2$$

$$= 0 + 0 + \frac{1}{2}(5.00\text{ rad/s}^2)(1.60\text{ s})^2 = 6.40\text{ rad}$$

The length pulled from the spool is

$$s = r\theta = (0.500\text{ m})(6.40\text{ rad}) = 3.20\text{ m}$$

When the spool reaches an angular velocity of 8.00 rad/s, 1.60 s will have elapsed and 3.20 m of cord will have been removed from the spool. Remaining on the spool will be $\boxed{0.800\text{ m}}$.

P10.68 (a) We consider the elevator-sheave-counterweight-Earth system, including n passengers, as an isolated system and apply the conservation of mechanical energy. We take the initial configuration, at the moment the drive mechanism switches off, as representing zero gravitational potential energy of the system.

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Therefore, the initial mechanical energy of the system [elevator (e), counterweight (c), sheave (s)] is

$$E_i = K_i + U_i = \frac{1}{2}m_e v^2 + \frac{1}{2}m_c v^2 + \frac{1}{2}I_s\omega^2 + 0$$
$$= \frac{1}{2}m_e v^2 + \frac{1}{2}m_c v^2 + \frac{1}{2}\left[\frac{1}{2}m_s r^2\right]\left(\frac{v}{r}\right)^2$$
$$= \frac{1}{2}\left[m_e + m_c + \frac{1}{2}m_s\right]v^2$$

The final mechanical energy of the system is entirely gravitational because the system is momentarily at rest:

$$E_f = K_f + U_f = 0 + m_e g d - m_c g d$$

where we have recognized that the elevator car goes up by the same distance d that the counterweight goes down. Setting the initial and final energies of the system equal to each other, we have

$$\frac{1}{2}\left[m_e + m_c + \frac{1}{2}m_s\right]v^2 = \left(m_e - m_c\right)gd$$

$$\frac{1}{2}\left\{\left[800\text{ kg} + n\left(80.0\text{ kg}\right)\right] + 950\text{ kg} + 140\text{ kg}\right\}\left(3.00\text{ m/s}\right)^2$$
$$= \left[800\text{ kg} + n\left(80.0\text{ kg}\right) - 950\text{ kg}\right](9.80\text{ m/s}^2)d$$

$$\boxed{d = (1890 + 80n)\left(\frac{0.459\text{ m}}{80n - 150}\right)}$$

(b) For $n = 2$: $d = (1890 + 80.0\times 2)\dfrac{0.459\text{ m}}{(80.0\times 2 - 150)} = \boxed{94.1\text{ m}}$

(c) For $n = 12$: $d = (1890 + 80.0\times 12)\dfrac{0.459\text{ m}}{(80.0\times 12 - 150)} = \boxed{1.62\text{ m}}$

(d) For $n = 0$: $d = (1890 + 80.0\times 0)\dfrac{0.459\text{ m}}{(80.0\times 0 - 150)} = \boxed{-5.79\text{ m}}$

(e) The raising car will coast to a stop only for $n \geq 2$.

(f) For $n = 0$ or $n = 1$, the mass of the elevator is less than the counterweight, so the car would accelerate upward if released.

(g) For $n \to \infty$, $d \to 80n\left(0.459\text{ m}\right)/(80n) = \boxed{0.459\text{ m}}$

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P10.69 (a) We find the angular speed by integrating the angular acceleration, which is given as $\alpha = -10.0 - 5.00t = \dfrac{d\omega}{dt}$, where α is in rad/s^2 and t is in seconds:

$$\Delta\omega = \int_{65.0}^{\omega} d\omega = \int_{0}^{t} [-10.0 - 5.00t]\,dt$$

$$\omega - 65.0 = -10.0t - 2.50t^2 \rightarrow \omega = 65.0 - 10.0t - 2.50t^2$$

where ω is in rad/s and t is in seconds.

For $t = 3.00$ s: $\omega = 65.0 - 10.0(3.00) - 2.50(3.00)^2 = \boxed{12.5\,\text{rad/s.}}$

(b) $\omega = \dfrac{d\theta}{dt} = 65.0 \text{ rad/s} - (10.0 \text{ rad/s}^2)t - (2.50 \text{ rad/s}^3)t^2$

Suppressing units,

$$\Delta\theta = \int_{0}^{t} \omega\,dt = \int_{0}^{t} \left[65.0 - 10.0t - 2.50t^2\right]dt$$

$$\Delta\theta = 65.0t - 5.00t^2 - (2.50/3)t^3$$

$$\Delta\theta = 65.0t - 5.00t^2 - 0.833t^3$$

At $t = 3.00$ s,

$$\Delta\theta = (65.0 \text{ rad/s})(3.00 \text{ s}) - (5.00 \text{ rad/s}^2)(9.00 \text{ s}^2) - (0.833 \text{ rad/s}^3)(27.0 \text{ s}^3)$$

$$\Delta\theta = \boxed{128 \text{ rad}}$$

P10.70 (a) We find the angular speed by integrating the angular acceleration, which is given as $\alpha(t) = A + Bt = \dfrac{d\omega}{dt}$, where the shaft is turning at angular speed ω at time $t = 0$.

$$\Delta\omega = \int_{\omega(0)}^{\omega(t)} d\omega = \int_{0}^{t} [A + Bt]\,dt$$

$$\omega(t) - \omega(0) = At + \frac{1}{2}Bt^2, \text{ and } \omega(0) = \omega \rightarrow \boxed{\omega(t) = \omega + At + \frac{1}{2}Bt^2}$$

(b) $\dfrac{d\theta}{dt} = \omega + At + \dfrac{1}{2}Bt^2$

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$$\Delta\theta = \int_0^t \omega(t)\,dt = \int_0^t \left[\omega + At + \frac{1}{2}Bt^2\right] dt$$

$$\Delta\theta = \boxed{\omega t + \frac{1}{2}At^2 + \frac{1}{6}Bt^3}$$

***P10.71** The resistive force on each ball is $R = D\rho A v^2$. Here $v = r\omega$, where r is the radius of each ball's path. The resistive torque on each ball is $\tau = rR$, so the total resistive torque on the three-ball system is $\tau_{\text{total}} = 3rR$.

The power required to maintain a constant rotation rate is $P = \tau_{\text{total}}\omega = 3rR\omega$. This required power may be written as

$$P = \tau_{\text{total}}\omega = 3r\left[D\rho A(r\omega)^2\right]\omega = \left(3r^3 DA\omega^3\right)\rho$$

with

$$\omega = \frac{2\pi \text{ rad}}{1 \text{ rev}}\left(\frac{10^3 \text{ rev}}{1 \text{ min}}\right)\left(\frac{1 \text{ min}}{60.0 \text{ s}}\right) = \frac{1\,000\pi}{30.0} \text{ rad/s}$$

Then

$$P = 3(0.100 \text{ m})^3(0.600)\left(4.00\times10^{-4} \text{ m}^2\right)\left(\frac{1\,000\pi}{30.0 \text{ s}}\right)^3 \rho$$

or $P = \left(0.827 \text{ m}^5/\text{s}^3\right)\rho$, where ρ is the density of the resisting medium.

(a) In air, $\rho = 1.20 \text{ kg/m}^3$, and

$$P = \left(0.827 \text{ m}^5/\text{s}^3\right)\left(1.20 \text{ kg/m}^3\right) = 0.992 \text{ N}\cdot\text{m/s} = \boxed{0.992 \text{ W}}$$

(b) In water, $\rho = 1\,000 \text{ kg/m}^3$ and $P = \boxed{827 \text{ W}}$.

***P10.72** Consider the total weight of each hand to act at the center of gravity (midpoint) of that hand. Then the total torque (taking CCW as positive) of these hands about the center of the clock is given by

$$\tau = -m_h g\left(\frac{L_h}{2}\right)\sin\theta_h - m_m g\left(\frac{L_m}{2}\right)\sin\theta_m$$

$$= -\frac{g}{2}\left(m_h L_h \sin\theta_h + m_m L_m \sin\theta_m\right)$$

If we take $t = 0$ at 12 o'clock, then the angular positions of the hands at time t are $\theta_h = \omega_h t$, where $\omega_h = \frac{\pi}{6}$ rad/h and $\theta_m = \omega_m t$, where $\omega_m = 2\pi$ rad/h. Therefore,

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$$\tau = \left(-4.90\ \text{m/s}^2\right) \times \left[(60.0\ \text{kg})(2.70\ \text{m})\sin\left(\frac{\pi t}{6}\right) + (100\ \text{kg})(4.50\ \text{m})\sin 2\pi t\right]$$

or $$\tau = (-794\ \text{N}\cdot\text{m})\left[\sin\left(\frac{\pi t}{6}\right) + 2.78\sin 2\pi t\right], \text{ where } t \text{ is in hours.}$$

(a) (i) At 3:00, $t = 3.00$ h, so

$$\tau = (-794\ \text{N}\cdot\text{m})\left[\sin\left(\frac{\pi}{2}\right) + 2.78\sin 6\pi\right] = \boxed{-794\ \text{N}\cdot\text{m}}$$

(ii) At 5:15, $t = 5\ \text{h} + \frac{15}{60}\ \text{h} = 5.25\ \text{h},$ and substitution gives:

$$\tau = \boxed{-2\,510\ \text{N}\cdot\text{m}}$$

(iii) At 6:00, $\tau = \boxed{0\ \text{N}\cdot\text{m}}$

(iv) At 8:20, $\tau = \boxed{-1\,160\ \text{N}\cdot\text{m}}$

(v) At 9:45, $\tau = \boxed{2\,940\ \text{N}\cdot\text{m}}$

(b) The total torque is zero at those times when

$$\sin\left(\frac{\pi t}{6}\right) + 2.78\sin 2\pi t = 0$$

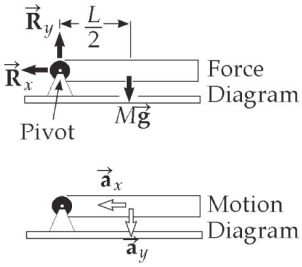
We proceed numerically, to find 0, 0.515 295 5, ..., corresponding to the times

12:00:00	12:30:55	12:58:19	1:32:31	1:57:01
2:33:25	2:56:29	3:33:22	3:56:55	4:32:24
4:58:14	5:30:52	6:00:00	6:29:08	7:01:46
7:27:36	8:03:05	8:26:38	9:03:31	9:26:35
10:02:59	10:27:29	11:01:41	11:29:05	

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P10.73 (a) Since only conservative forces are acting on the bar, we have conservation of energy of the bar-Earth system:

$$K_i + U_i = K_f + U_f$$



ANS. FIG. P10.73

For evaluation of the gravitational energy of the system, a rigid body can be modeled as a particle at its center of mass. Take the zero configuration for potential energy for the bar-Earth system with the bar horizontal.

Under these conditions, $U_f = 0$ and $U_i = MgL/2$.

Using the conservation of energy equation above,

$$0 + \frac{1}{2}MgL = \frac{1}{2}I\omega_f^2 \quad \text{and} \quad \omega_f = \sqrt{MgL/I}$$

For a bar rotating about an axis through one end, $I = ML^2/3$.

Therefore,

$$\omega_f = \sqrt{\frac{MgL}{\frac{1}{3}ML^2}} = \boxed{\sqrt{\frac{3g}{L}}}$$

Note that we have chosen clockwise rotation as positive.

(b) $\sum \tau = I\alpha$: $\quad Mg\left(\frac{L}{2}\right) = \left(\frac{1}{3}ML^2\right)\alpha$ and $\alpha = \boxed{\frac{3g}{2L}}$

(c) $$a_x = -a_c = -r\omega_f^2 = -\left(\frac{L}{2}\right)\left(\frac{3g}{L}\right) = -\frac{3g}{2}$$

Since this is **centripetal** acceleration, it is directed along the **negative** horizontal.

$$a_y = -a_t = -r\alpha = \frac{L}{2}\alpha = -\frac{3g}{4}$$

$$\boxed{\vec{\mathbf{a}} = -\frac{3}{2}g\hat{\mathbf{i}} - \frac{3}{4}g\hat{\mathbf{j}}}$$

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(d) The pivot exerts a force $\vec{\mathbf{F}}$ on the rod. Using Newton's second law, we find

$$F_x = Ma_x = -\frac{3}{2}Mg$$

$$F_y - Mg = Ma_y = -\frac{3}{4}Mg \rightarrow F_y = Mg - \frac{3}{4}Mg = \frac{1}{4}Mg$$

$$\boxed{\vec{\mathbf{F}} = M\vec{\mathbf{a}} = -\frac{3}{2}Mg\hat{\mathbf{i}} + \frac{1}{4}Mg\hat{\mathbf{j}}}$$

P10.74 We assume that air resistance has a negligible effect on a drop so that mechanical energy is conserved in the drop-Earth system. The first drop leaving the wheel has a velocity v_1 directed upward. The magnitude of this velocity is found from

$$K_i + U_{gi} = K_f + U_{gf}$$

$$\frac{1}{2}mv_1^2 + 0 = 0 + mgh_1$$

so $$v_1 = \sqrt{2gh_1} = \sqrt{2\left(9.80\ \text{m/s}^2\right)(0.540\ \text{m})} = 3.25\ \text{m/s}$$

Similarly, the second drop has a velocity given by

$$v_2 = \sqrt{2gh_2} = \sqrt{2\left(9.80\ \text{m/s}^2\right)(0.510\ \text{m})} = 3.16\ \text{m/s}$$

From $\omega = \frac{v}{r}$, we find

$$\omega_1 = \frac{v_1}{r} = \frac{3.25\ \text{m/s}}{0.381\ \text{m}} = 8.53\ \text{rad/s}$$

and $$\omega_2 = \frac{v_2}{r} = \frac{3.16\ \text{m/s}}{0.381\ \text{m}} = 8.29\ \text{rad/s}$$

or $$\alpha = \frac{\omega_2^2 - \omega_1^2}{2\Delta\theta} = \frac{\left(8.29\ \text{rad/s}\right)^2 - \left(8.53\ \text{rad/s}\right)^2}{4\pi} = \boxed{-0.322\ \text{rad/s}^2}$$

P10.75 We assume that air resistance has a negligible effect on a drop so that mechanical energy is conserved in the drop-Earth system. At the instant it comes off the wheel, the first drop has a velocity v_1 directed upward. The magnitude of this velocity is found from

$$K_i + U_{gi} = K_f + U_{gf}$$

$$\frac{1}{2}mv_1^2 + 0 = 0 + mgh_1 \text{ or } v_1 = \sqrt{2gh_1}$$

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The angular velocity of the wheel at the instant the first drop leaves is

$$\omega_1 = \frac{v_1}{R} = \sqrt{\frac{2gh_1}{R^2}}$$

Similarly for the second drop: $v_2 = \sqrt{2gh_2}$ and $\omega_2 = \frac{v_2}{R} = \sqrt{\frac{2gh_2}{R^2}}$

The angular acceleration of the wheel is then

$$a = \frac{\omega_2^2 - \omega_1^2}{2\Delta\theta} = \frac{2gh_2/R^2 - 2gh_1/R^2}{2(2\pi)} = \boxed{\frac{g(h_2 - h_1)}{2\pi R^2}}$$

P10.76 (a) Modeling the Earth as a sphere, its rotational kinetic energy is

$$K = \frac{1}{2}\left(\frac{2}{5}MR^2\right)(\omega^2)$$

$$= \frac{1}{2}\left[\frac{2}{5}(5.98\times10^{24}\ \text{kg})(6.37\times10^{6}\ \text{m})^2\right]\left(\frac{2\pi}{86\,400\ \text{s}}\right)^2$$

$$= \boxed{2.57\times10^{29}\ \text{J}}$$

(b) The change in rotational kinetic energy is found by differentiating the equation for rotational kinetic energy with respect to time:

$$\frac{dK}{dt} = \frac{d}{dt}\left[\frac{1}{2}\left(\frac{2}{5}MR^2\right)\left(\frac{2\pi}{T}\right)^2\right]$$

$$= \frac{1}{5}MR^2(2\pi)^2(-2T^{-3})\frac{dT}{dt}$$

$$= \frac{1}{5}MR^2\left(\frac{2\pi}{T}\right)^2\left(\frac{-2}{T}\right)\frac{dT}{dt} = K\left(\frac{-2}{T}\right)\frac{dT}{dt}$$

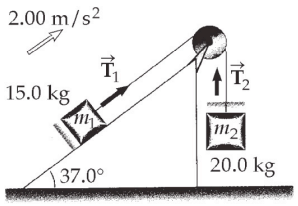
Substituting,

$$\frac{dK}{dt} = (2.57\times10^{29}\ \text{J})\left(\frac{-2}{86\,400\ \text{s}}\right)\left(\frac{10\times10^{-6}\ \text{s}}{3.16\times10^{7}\ \text{s}}\right)(86\,400\ \text{s/day})$$

$$= \boxed{-1.63\times10^{17}\ \text{J/day}}$$

P10.77 (a) We apply the **particle under a net force** model to each block.

(b) We apply the **rigid object under a net torque** model to the pulley.



ANS. FIG. P10.77

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(c) We use $\sum F = ma$ for each block to find each string tension. The forces acting on the 15-kg block are its weight, the normal support from the incline, and T_1. Taking the positive x axis as directed up the incline,

$$\sum F_x = ma_x \quad \text{yields:} \quad -(m_1 g)_x + T_1 = m_1(+a)$$

Solving and substituting known values, we have

$$\begin{aligned} T_1 &= m_1(+a) + (m_1 g)_x \\ &= (15.0\ \text{kg})(2.00\ \text{m/s}^2) + (15.0\ \text{kg})(9.80\ \text{m/s}^2)\sin 37.0° \\ &= \boxed{118\ \text{N}} \end{aligned}$$

(d) Similarly, for the counterweight, we have

$$\sum F_y = ma_y \quad \text{or} \quad T_2 - m_2 g = m_2(-a)$$

$$\begin{aligned} T_2 &= m_2 g + m_2(-a) \\ &= (20.0\ \text{kg})(9.80\ \text{m/s}^2) + (20.0\ \text{kg})(-2.00\ \text{m/s}^2) \\ &= \boxed{156\ \text{N}} \end{aligned}$$

(e) Now for the pulley,

$$\sum \tau = r(T_2 - T_1) = I\alpha = I\,a/r$$

so $$I = \frac{r^2}{a}(T_2 - T_1)$$

where we have chosen to call clockwise positive.

(f) Computing from above, the pulley's rotational inertia is

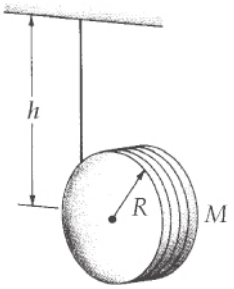
$$I = \frac{r^2}{a}(T_2 - T_1) = \frac{(156\ \text{N} - 118\ \text{N})(0.250\ \text{m})^2}{2.00\ \text{m/s}^2} = \boxed{1.17\ \text{kg}\cdot\text{m}^2}$$

P10.78 Choosing positive linear quantities to be downwards and positive angular quantities to be clockwise, $\sum F_y = ma_y$ yields

$$\sum F = Mg - TM = a \quad \text{or} \quad a = \frac{Mg - T}{M}$$

$\sum \tau = I\alpha$ then becomes

$$\sum \tau = TR = I\alpha = \frac{1}{2}MR^2\left(\frac{a}{R}\right) \quad \text{so} \quad a = \frac{2T}{M}$$



ANS. FIG. P10.78

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(a) Setting these two expressions equal,

$$\frac{Mg - T}{M} = \frac{2T}{M} \quad \text{and} \quad T = \boxed{Mg/3}$$

(b) Substituting back,

$$a = \frac{2T}{M} = \frac{2Mg}{3M} \quad \text{or} \quad a = \boxed{\frac{2}{3}g}$$

(c) Since $v_i = 0$ and $a = \frac{2}{3}g$, $v_f^2 = v_i^2 + 2ah$ gives us $v_f^2 = 0 + 2\left(\frac{2}{3}g\right)h$,

or $\quad v_f = \boxed{\sqrt{4gh/3}}$

(d) Now we verify this answer. Requiring conservation of mechanical energy for the disk-Earth system, we have

$$U_i + K_{\text{rot},\,i} + K_{\text{trans},\,i} = U_f + K_{\text{rot},\,f} + K_{\text{trans},\,f}$$

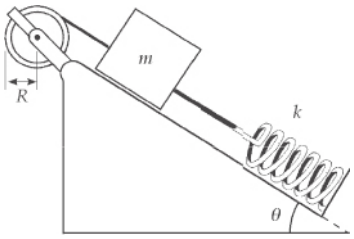
$$mgh + 0 + 0 = 0 + \frac{1}{2}I\omega^2 + \frac{1}{2}mv^2$$

$$mgh = \frac{1}{2}\left(\frac{1}{2}MR^2\right)\omega^2 + \frac{1}{2}Mv^2$$

When there is no slipping, $\omega = \frac{v}{R}$ and $v = \sqrt{\frac{4gh}{3}}$.

The answer is the same.

P10.79 The block and end of the spring are pulled a distance d up the incline and then released. The angular speed of the reel and the speed of the block are related by $v = \omega R$. The block-reel-Earth system is isolated, so



ANS. FIG. P10.79

$$\Delta K + \Delta U = 0 \rightarrow K_f - K_i + U_f - U_i = 0$$

$$\left(\frac{1}{2}mv^2 - 0\right) + \left(\frac{1}{2}I\omega^2 - 0\right)$$

$$+ \left(0 - mgd\sin\theta\right) + \left(0 - \frac{1}{2}kd^2\right) = 0$$

$$\frac{1}{2}\omega^2\left(I + mR^2\right) = mgd\sin\theta + \frac{1}{2}kd^2$$

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$$\boxed{\omega = \sqrt{\frac{2mgd\sin\theta + kd^2}{I + mR^2}}}$$

P10.80 The center of gravity of the uniform board is at its middle. For the board just starting to move,

$$\sum \tau = I\alpha:$$

$$mg\left(\frac{\ell}{2}\right)\cos\theta = \left(\frac{1}{3}m\ell^2\right)\alpha$$

$$\alpha = \frac{3}{2}\left(\frac{g}{\ell}\right)\cos\theta$$



ANS. FIG. P10.80

The tangential acceleration of the end is $a_t = \ell\alpha = \frac{3}{2}g\cos\theta$

and its vertical component is $a_y = a_t\cos\theta = \frac{3}{2}g\cos^2\theta.$

If this is greater than g, the board will pull ahead of the falling ball:

(a) $\frac{3}{2}g\cos^2\theta \geq g$ gives $\cos^2\theta \geq \frac{2}{3}$ so $\cos\theta \geq \sqrt{\frac{2}{3}}$ and $\boxed{\theta \leq 35.3°}$

(b) When $\theta = 35.3°$, the cup will land underneath the release point of the ball if $r_c = \ell\cos\theta$.

When $\ell = 1.00\text{ m}$ and $\theta = 35.3°$,

$$r_c = 1.00\text{ m}\sqrt{\frac{2}{3}} = 0.816\text{ m}$$

so the cup should be

$$\ell - r_c = 1.00\text{ m} - 0.816\text{ m} = \boxed{0.184\text{ m from the moving end}}$$

P10.81 For the isolated sphere-Earth system, energy is conserved,

so

$$\Delta U + \Delta K_{\text{rot}} + \Delta K_{\text{trans}} = 0$$

$$mg\left(R - r\right)\left(\cos\theta - 1\right) + \left[\frac{1}{2}mv^2 - 0\right] + \frac{1}{2}\left[\frac{2}{5}mr^2\right]\omega^2 = 0$$

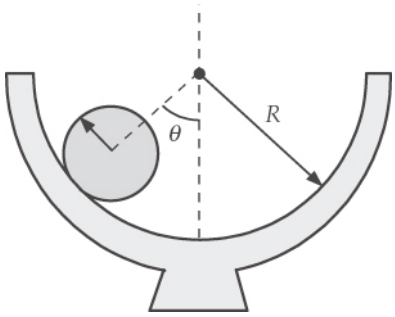
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Substituting $v = r\omega$, we obtain

$$mg(R-r)(\cos\theta - 1) + \left[\frac{1}{2}m(r\omega)^2 - 0\right] + \frac{1}{2}\left[\frac{2}{5}mr^2\right]\omega^2 = 0$$

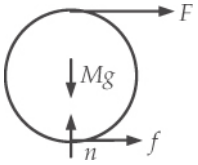
$$mg(R-r)(\cos\theta - 1) + \left[\frac{1}{2} + \frac{1}{5}\right]mr^2\omega^2 = 0$$

$$\boxed{\omega = \sqrt{\left(\frac{10}{7}\right)\frac{(R-r)(1-\cos\theta)g}{r^2}}}$$



ANS. FIG. P10.81

P10.82 (a) From the particle under a net force model in the x direction, we have



ANS. FIG. P10.82

$$\sum F_x = F + f = Ma_{CM}$$

From the particle under a net torque model,

$$\sum \tau = FR - fR = I\alpha$$

Combining the two equations, and noting that $I = \frac{1}{2}MR^2$, gives

$$FR - \left(Ma_{CM} - F\right)R = \frac{Ia_{CM}}{R} \qquad \boxed{a_{CM} = \frac{4F}{3M}}$$

(b) Assuming friction is to the right, then

$$f + F = Ma_{CM} = M\left(\frac{4F}{3M}\right)$$

$$\rightarrow f = M\left(\frac{4F}{3M}\right) - F = \boxed{\frac{1}{3}F}$$

The facts that (1) we assumed that friction is to the right in Figure P10.82 and (2) our value for f comes out positive indicate that the friction force must indeed be to the right.

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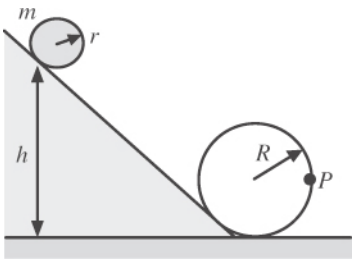
(c) From the kinematic equations,

$$v_f^2 = v_i^2 + 2a\left(x_f - x_i\right)$$
$$= 0 + 2ad$$

or

$$v_f = \sqrt{2ad} = \boxed{\sqrt{\frac{8Fd}{3M}}}$$

P10.83 (a) $\Delta K_{\text{rot}} + \Delta K_{\text{trans}} + \Delta U = 0$



ANS. FIG. P10.83

Note that initially the center of mass of the sphere is slightly higher than the distance h above the bottom of the loop; and as the mass reaches the top of the loop, this distance above the reference level is $2R - r$, but we are told that $r << R$, so we ignore r when considering heights for the gravitational potential energy of the sphere-Earth system. The conservation of energy requirement gives

$$mgh = mg\left(2R\right) + \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

For the sphere $I = \frac{2}{5}mr^2$ and $v = r\omega$, so that the expression becomes

$$gh = 2gR + \frac{7}{10}v^2 \qquad \textbf{[1]}$$

Note that $h = h_{\text{min}}$ when the speed of the sphere at the top of the loop satisfies the condition

$$\sum F = mg = \frac{mv^2}{R} \quad \text{or} \quad v^2 = gR$$

Substituting this into equation [1] gives

$$h_{\text{min}} = 2R + 0.700R \text{ or } \boxed{h_{\text{min}} = 2.70R}$$

(b) When the sphere is initially at $h = 3R$ and finally at point P, the conservation of energy equation gives

$$mg3R = mgR + \frac{1}{2}mv^2 + \frac{1}{5}mv^2, \text{ or } v^2 = \frac{20}{7}Rg$$

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Turning clockwise as it rolls without slipping past point P, the sphere is slowing down with counterclockwise angular acceleration caused by the torque of an upward force f of static friction. We have

$$\sum F_y = ma_y \quad \rightarrow \quad f - mg = -m\alpha r$$

and $\sum \tau = I\alpha \rightarrow fr = \left(\frac{2}{5}\right)mr^2\alpha.$

Eliminating f by substitution yields

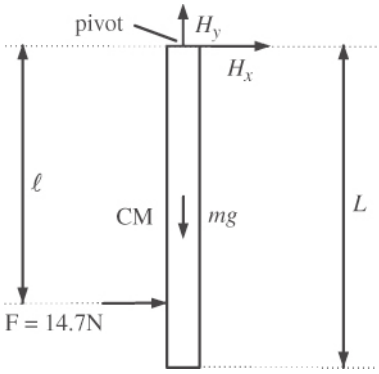
$$\alpha = \frac{5g}{7r} \text{ so that } \sum F_y = \boxed{-\frac{5}{7}mg}$$

P10.84 The length of the rod is L, and the horizontal force is applied the vertical distance L from the hinge. Consider the free-body diagram shown. The sum of torques about the chosen pivot is

$$\sum \tau = I\alpha \Rightarrow F = \left(\frac{1}{3}mL^2\right)\left(\frac{a_{\text{CM}}}{\text{L}/2}\right) = \left(\frac{2}{3}mL\right)a_{\text{CM}} \qquad \textbf{[1]}$$

(a) $\ell = L = 1.24$ m: In this case, equation [1] becomes

$$a_{\text{CM}} = \frac{3F}{2m} = \frac{3(14.7\text{ N})}{2(0.630\text{ kg})} = \boxed{35.0\text{ m/s}^2}$$



ANS. FIG. P10.84

(b) We apply the particle under a net force model in the horizontal direction (see ANS. FIG. P10.84 for the labelling of forces):

$$\sum F_x = ma_{\text{CM}} \Rightarrow F + H_x = ma_{\text{CM}}$$

or $H_x = ma_{\text{CM}} - F$

Thus,

$$H_x = (0.630\text{ kg})(35.0\text{ m/s}^2) - 14.7\text{ N} = +7.35\text{ N}$$

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or $\vec{\mathbf{H}}_x = \boxed{7.35\hat{\mathbf{i}}\ \text{N}}$

(c) With $\ell = \frac{1}{2}L = 0.620\ \text{m}$, equation [1] yields

$$a_{\text{CM}} = \frac{3F}{4m} = \frac{3(14.7\ \text{N})}{4(0.630\ \text{kg})} = \boxed{17.5\ \text{m/s}^2}$$

(d) Again, $\sum F_x = ma_{\text{CM}} \Rightarrow H_x = ma_{\text{CM}} - F$, so

$$H_x = (0.630\ \text{kg})(17.5\ \text{m/s}^2) - 14.7\ \text{N} = -3.68\ \text{N}$$

or $\mathbf{H}_x = \boxed{-3.68\hat{\mathbf{i}}\ \text{N}}$

(e) If $H_x = 0$, then $\sum F_x = ma_{\text{CM}} \Rightarrow F = ma_{\text{CM}}$, or $a_{\text{CM}} = \frac{F}{m}$.

Thus, equation [1] becomes

$$F\ell = \left(\frac{2}{3}mL\right)\left(\frac{F}{m}\right)$$

so $\ell = \frac{2}{3}L = \frac{2}{3}(1.24\ \text{m}) = \boxed{0.827\ \text{m (from the top)}}$

P10.85 Note that when the CM of the falling rod is very near the surface, the velocity of the end of the rod in contact with the surface is a combination of the downward motion of the CM and the upward motion of the rotating end: $v_{\text{end}} = v_{\text{CM}} - \omega r$. Because the velocity of this end relative to the surface is zero,

$$v_{\text{end}} = v_{\text{CM}} - \omega(h/2) = 0 \rightarrow v_{\text{CM}} = \omega(h/2)t$$

(a) There are no horizontal forces acting on the rod, so the center of mass (CM) will not move horizontally. Rather, the center of mass drops straight downward (distance $h/2$) with the rod rotating about the center of mass as it falls.

From conservation of energy:

$$K_f + U_{gf} = K_i + U_{gi}$$

$$\frac{1}{2}Mv_{\text{CM}}^2 + \frac{1}{2}I\omega^2 + 0 = 0 + Mg\left(\frac{h}{2}\right) \text{ or}$$

$$\frac{1}{2}Mv_{\text{CM}}^2 + \frac{1}{2}\left(\frac{1}{12}Mh^2\right)\left(\frac{v_{\text{CM}}}{\text{h}/2}\right)^2 = Mg\left(\frac{h}{2}\right)$$

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which reduces to

$$v_{\text{CM}} = \boxed{\sqrt{\frac{3gh}{4}}}$$

(b) In this case, the motion is a pure rotation about a fixed pivot point (the lower end of the rod) with the center of mass moving in a circular path of radius $h/2$. From conservation of energy:

$$K_f + U_{gf} = K_i + U_{gi}$$

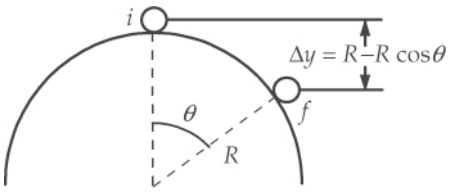
$$\frac{1}{2}I\omega^2 + 0 = 0 + Mg\left(\frac{h}{2}\right) \text{ or}$$

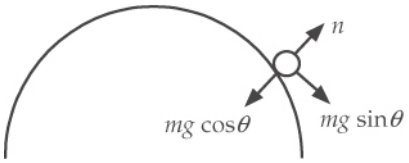
$$\frac{1}{2}\left(\frac{1}{3}Mh^2\right)\left(\frac{v_{\text{CM}}}{\text{h}/2}\right)^2 = Mg\left(\frac{h}{2}\right)$$

which reduces to

$$v_{\text{CM}} = \boxed{\sqrt{\frac{3gh}{4}}}$$

P10.86 The grape-Earth system is isolated, so mechanical energy in that system is conserved. Between top of the clown's head and the point where the grape leaves the surface:





ANS. FIG. P10.86

$$K_i + U_i = K_f + U_f$$

$$0 + mg\Delta y = \frac{1}{2}mv_f^2 + \frac{1}{2}I\omega_f^2 + 0$$

$$mgR(1-\cos\theta) = \frac{1}{2}mv_f^2 + \frac{1}{2}\left(\frac{2}{5}mR^2\right)\left(\frac{v_f}{R}\right)^2$$

which gives

$$g\left(1-\cos\theta\right) = \frac{7}{10}\left(\frac{v_f^2}{R}\right) \qquad \textbf{[1]}$$

Consider the radial forces acting on the grape:

$$mg\cos\theta - n = \frac{mv_f^2}{R}$$

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At the point where the grape leaves the surface, $n \to 0$. Thus,

$$mg\cos\theta = \frac{mv_f^2}{R} \quad \text{or} \quad \frac{v_f^2}{R} = g\cos\theta$$

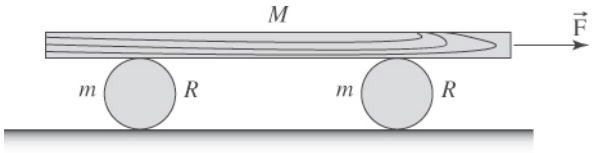
Substituting this into equation [1] gives

$$g - g\cos\theta = \frac{7}{10}g\cos\theta$$

or $$\theta = \cos^{-1}\left(\frac{10}{17}\right) = \boxed{54.0°}$$

Challenge Problems

P10.87 Refer to the force diagrams for the plank and rollers in ANS. FIG. P10.87(b) below. Call f_t the frictional force exerted by each roller backward on the plank. Name as f_b the rolling resistance exerted backward by the ground on each roller.



ANS. FIG. P10.87(a)

For the plank,

$$\sum F_x = ma_x: \quad 6.00\,\text{N} - 2f_t = (6.00\,\text{kg})a_p \qquad \textbf{[1]}$$

If we think of the motion of a roller as a small rotation about its point of contact with the surface, we see that the center of each roller moves forward only half as far as the plank.

Each roller has acceleration $\frac{a_p}{2}$ and angular acceleration

$$\frac{a_p/2}{5.00\text{ cm}} = \frac{a_p}{0.100\text{ m}}$$

Then for each,

$$\sum F_x = ma_x: \quad +f_t - f_b = (2.00\,\text{kg})\frac{a_p}{2} \qquad \textbf{[2]}$$

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$$\sum \tau = I\alpha:$$

$$f_t(5.00\text{ cm}) + f_b(5.00\text{ cm}) = \frac{1}{2}(2.00\text{ kg})(5.00\text{ cm})^2 \frac{a_p}{10.0\text{ cm}}$$

So $$f_t + f_b = \left(\frac{1}{2}\text{ kg}\right)a_p \qquad \textbf{[3]}$$

Add equations [2] and [3] to eliminate f_b: $2f_t = (1.50\text{ kg})a_p$

(a) Substituting the value for $2f_t$ into equation [1] gives

$$6.00\text{ N} - (1.50\text{ kg})a_p = (6.00\text{ kg})a_p$$

$$\rightarrow a_p = \frac{6.00\text{ N}}{7.50\text{ kg}} = \boxed{0.800\text{ m/s}^2}$$

(b) For each roller, $a = \frac{a_p}{2} = \boxed{0.400\text{ m/s}^2}$

(c) Substituting back,

$$2f_t = (1.50\text{ kg})(0.800\text{ m/s}^2)$$

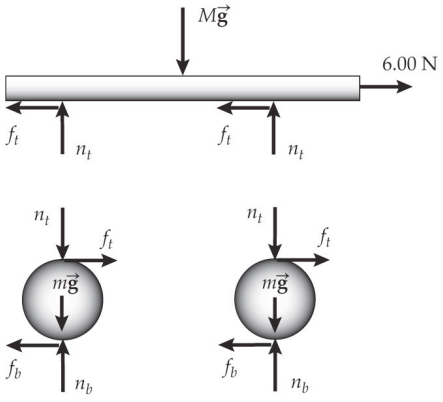
$$f_t = \boxed{0.600\text{ N}}$$

then, from equation [3],

$$0.600\text{ N} + f_b = \left(\frac{1}{2}\text{ kg}\right)(0.800\text{ m/s}^2)$$

$$f_b = -0.200\text{ N}$$

The negative sign means that the horizontal force of ground on each roller is $\boxed{0.200\text{ N forward}}$ rather than backward as we assumed.



ANS. FIG. P10.87(b)

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P10.88 For large energy storage at a particular rotation rate, we want a large moment of inertia. To combine this requirement with small mass, we place the mass as far away from the axis as possible.



ANS. FIG. P10.88

We choose to make the flywheel as a hollow cylinder 18.0 cm in diameter and 8.00 cm long. To support this rim, we place a disk across its center. We assume that a disk 2.00 cm thick will be sturdy enough to support the hollow cylinder securely.

The one remaining adjustable parameter is the thickness of the wall of the hollow cylinder. From Table 10.2, the moment of inertia can be written as

$$I_{\text{disk}} + I_{\text{hollow cylinder}} = \frac{1}{2}M_{\text{disk}}R_{\text{disk}}^2 + \frac{1}{2}M_{\text{wall}}\left(R_{\text{outer}}^2 + R_{\text{inner}}^2\right)$$

$$= \frac{1}{2}\rho V_{\text{disk}}R_{\text{outer}}^2 + \frac{1}{2}\rho V_{\text{wall}}\left(R_{\text{outer}}^2 + R_{\text{inner}}^2\right)$$

$$= \frac{\rho}{2}\pi R_{\text{outer}}^2(2.00\text{ cm})R_{\text{outer}}^2 + \frac{\rho}{2}\left[\pi R_{\text{outer}}^2 - \pi R_{\text{inner}}^2\right] \times(6.00\text{ cm})\left(R_{\text{outer}}^2 + R_{\text{inner}}^2\right)$$

$$= \frac{\rho\pi}{2}\left[(9.00\text{ cm})^4(2.00\text{ cm}) + (6.00\text{ cm})\left[(9.00\text{ cm})^2 - R_{\text{inner}}^2\right]\left[(9.00\text{ cm})^2 + R_{\text{inner}}^2\right]\right]$$

$$= \rho\pi\left[6\,561\text{ cm}^5 + (3.00\text{ cm})\left((9.00\text{ cm})^4 - R_{\text{inner}}^4\right)\right]$$

$$= \rho\pi\left[26\,244\text{ cm}^5 - (3.00\text{ cm})R_{\text{inner}}^4\right]$$

For the required energy storage,

$$\frac{1}{2}I\omega_1^2 = \frac{1}{2}I\omega_2^2 + W_{\text{out}}$$

$$\frac{1}{2}I\left[(800\text{ rev/min})\left(\frac{2\pi\text{ rad}}{1\text{ rev}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)\right]^2 - \frac{1}{2}I\left[(600\text{ rev/min})\left(\frac{2\pi\text{ rad}}{60\text{ s}}\right)\right]^2$$

$$= 60.0\text{ J}$$

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$$I = \frac{60.0 \text{ J}}{1\,535/\text{s}^2}$$

$$= \left(7.85 \times 10^3 \text{ kg/m}^3\right)\pi\left[26\,244 \text{ cm}^5 - (3.00 \text{ cm})R_{\text{inner}}^4\right]$$

$$1.58 \times 10^{-5} \text{ m}^5\left(\frac{100 \text{ cm}}{1 \text{ m}}\right)^5 = 26\,244 \text{ cm}^5 - (3.00 \text{ cm})R_{\text{inner}}^4$$

$$R_{\text{inner}} = \left(\frac{26\,244 \text{ cm}^4 - 15\,827 \text{ cm}^4}{3.00}\right)^{1/4} = 7.68 \text{ cm}$$

The inner radius of the flywheel is 7.68 cm. The mass of the flywheel is then 7.27 kg, found as follows:

$$M_{\text{disk}} + M_{\text{wall}} = \rho\pi R_{\text{outer}}^2(2.00 \text{ cm}) + \rho\left[\pi R_{\text{outer}}^2 - \pi R_{\text{inner}}^2\right](6.00 \text{ cm})$$

$$= \left(7.86 \times 10^3 \text{ kg/m}^3\right)\pi \left[(0.090 \text{ m})^2(0.020 \text{ m}) + \left[(0.090 \text{ m})^2 - (0.076\,8 \text{ m})^2\right](0.060 \text{ m})\right]$$

$$= 7.27 \text{ kg}$$

If we made the thickness of the disk somewhat less than 2.00 cm and the inner radius of the cylindrical wall less than 7.68 cm to compensate, the mass could be a bit less than 7.27 kg.

The flywheel can be shaped like a cup or open barrel, 9.00 cm in outer radius and 7.68 cm in inner radius, with its wall 6 cm high, and with its bottom forming a disk 2.00 cm thick and 9.00 cm in radius. It is mounted to the crankshaft at the center of this disk and turns about its axis of symmetry. Its mass is 7.27 kg. If the disk were made somewhat thinner and the barrel wall thicker, the mass could be smaller.

P10.89 (a) At $t = 0$, $\omega = 3.50 \text{ rad/s} = \omega_0 e^0$. Thus, $\boxed{\omega_0 = 3.50 \text{ rad/s}}$.

At $t = 9.30$ s, $\omega = 2.00 \text{ rad/s} = \omega_0 e^{-\sigma(9.30s)}$.

We now calculate σ: To solve $\omega = \omega_0 e^{-\sigma t}$ for σ, we recall that the natural logarithm function is the inverse of the exponential function.

$\omega/\omega_0 = e^{-\sigma t}$ becomes $\ln(\omega/\omega_0) = -\sigma t$ or $\ln(\omega_0/\omega) = +\sigma t$

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so $$\sigma = \left(\frac{1}{t}\right)\ln(\omega_0/\omega) = \left(\frac{1}{9.30\text{ s}}\right)\ln\left(\frac{3.50}{2.00}\right) = \frac{0.560}{9.30\text{ s}} = \boxed{6.02\times10^{-2}\text{ s}^{-1}}$$

(b) At all times,

$$\alpha = \frac{d\omega}{dt} = \frac{d}{dt}\left[\omega_0 e^{-\sigma t}\right] = -\sigma\omega_0 e^{-\sigma t}$$

At $t = 3.00$ s,

$$\alpha = -(0.060\ 2\text{ s}^{-1})(3.50\text{ rad/s})e^{-0.181} = \boxed{-0.176\text{ rad/s}^2}$$

(c) From the given equation, we have $d\theta = \omega_0^{e^{-\sigma t}}dt$

and

$$\theta = \int_{0\text{ s}}^{2.50\text{ s}} \omega_0 e^{-\sigma t}dt = \frac{\omega_0}{-\sigma}e^{-\sigma t}\Big|_{0\text{ s}}^{2.50\text{ s}} = \frac{\omega_0}{-\sigma}\left(e^{-2.50\sigma} - 1\right)$$

Substituting and solving,

$$\theta = -58.2(0.860 - 1)\text{ rad} = 8.12\text{ rad}$$

or $$\theta = (8.12\text{ rad})\left(\frac{1\text{ rev}}{2\pi\text{ rad}}\right) = \boxed{1.29\text{ rev}}$$

(d) The motion continues to a finite limit, as ω approaches zero and t goes to infinity. From part (c), the total angular displacement is

$$\theta = \int_0^\infty \omega_0 e^{-\sigma t}dt = \frac{\omega_0}{-\sigma}e^{-\sigma t}\Big|_0^\infty = \frac{\omega_0}{-\sigma}(0 - 1) = \frac{\omega_0}{\sigma}$$

Substituting,

$$\theta = 58.2\text{ rad} \quad \text{or} \quad \theta = \left(\frac{1\text{ rev}}{2\pi\text{ rad}}\right)(58.2\text{ rad}) = \boxed{9.26\text{ rev}}$$

P10.90 (a) If we number the loops of the spiral track with an index n, with the innermost loop having $n = 0$, the radii of subsequent loops as we move outward on the disc is given by $r = r_i + hn$. Along a given radial line, each new loop is reached by rotating the disc through 2π rad. Therefore, the ratio $\theta/2\pi$ is the number of revolutions of the disc to get to a certain loop. This is also the number of that loop, so $n = \theta/2\pi$. Therefore, $r = r_i + h\,\theta/2\pi$.

(b) Starting from $\omega = v/r$, we substitute the definition of angular speed on the left and the result for r from part (a) on the right:

$$\omega = \frac{v}{r} \rightarrow \frac{d\theta}{dt} = \frac{v}{r_i + (h\theta/2\pi)}$$

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(c) Rearrange terms in preparation for integrating both sides:

$$\left(r_i + \frac{h}{2\pi}\theta\right)d\theta = v\,dt$$

and integrate from $\theta = 0$ to $\theta = \theta$ and from $t = 0$ to $t = t$:

$$r_i\theta + \frac{h}{4\pi}\theta^2 = vt$$

We rearrange this equation to form a standard quadratic equation in θ:

$$\frac{h}{4\pi}\theta^2 + r_i\theta - vt = 0$$

The solution to this equation is

$$\theta = \frac{-r_i \pm \sqrt{r_i^2 + \frac{h}{\pi}vt}}{\frac{h}{2\pi}} = \boxed{\frac{2\pi r_i}{h}\left(\sqrt{1 + \frac{vh}{\pi r_i^2}t} - 1\right)}$$

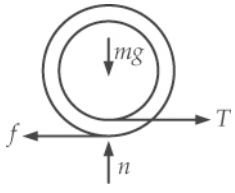
where we have chosen the positive root in order to make the angle θ positive.

(d) We differentiate the result in (c) twice with respect to time to find the angular acceleration, resulting in

$$\boxed{\alpha = -\frac{hv^2}{2\pi r_i^3\left(1 + \frac{vh}{\pi r_i^2}t\right)^{3/2}}}$$

Where we have used $\frac{d}{dx}\sqrt{u} = \frac{1}{2\sqrt{u}}\frac{du}{dx}$. Because this expression involves the time t, the angular acceleration is not constant.

P10.91 (a) $\sum F_x = ma_x$ reads $-f + T = ma$. If we take torques around the center of mass, we can use $\sum \tau = I\alpha$, which reads $+fR_2 - TR_1 = I\alpha$. For rolling without slipping, $\alpha = \frac{a}{R_2}$. By substitution,



ANS. FIG. P10.91

$$fR_2 - TR_1 = \frac{la}{R_2} = \frac{I}{R_2 m}(T - f)$$

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$$fR_2^2 m - TR_1R_2 m = IT - If$$
$$f\left(I + mR_2^2\right) = T\left(I + mR_1R_2\right)$$
$$f = \left(\frac{I + mR_1R_2}{I + mR_2^2}\right)T$$

(b) Since the answer is positive, the friction force is confirmed to be $\boxed{\text{to the left}}$.

P10.92 (a) From the isolated system model for the block-pulley-Earth system,

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0$$
$$\left(\frac{1}{2}Mv^2 - 0\right) + \left(\frac{1}{2}I\omega^2 - 0\right) + \left(0 - Mgd\sin\theta\right) + fd = 0$$
$$\frac{1}{2}Mv^2 + \frac{1}{2}\left(\frac{1}{2}mr^2\right)\left(\frac{v}{r}\right)^2 - Mgd\sin\theta + \left(\mu Mg\cos\theta\right)d = 0$$
$$v = \sqrt{\frac{4Mgd\left(\sin\theta - \mu\cos\theta\right)}{2M + m}}$$

(b) From the particle under constant acceleration model for the block,

$$v_f^2 = v_i^2 + 2ad$$
$$a = \frac{v_f^2 - v_i^2}{2d} = \frac{v^2}{2d} = \boxed{\frac{2Mg\left(\sin\theta - \mu\cos\theta\right)}{2M + m}}$$

P10.93 The location of the dog is described by $\theta_d = \left(0.750\ \text{rad/s}\right)t$. For the bone,

$$\theta_{\text{b}} = \frac{1}{3}2\pi\ \text{rad} + \frac{1}{2}0.015\ \text{rad/s}^2\ t^2$$

(a) We look for a solution to (suppressing units)

$$0.75t = \frac{2\pi}{3} + 0.007\ 5t^2$$
$$0 = 0.007\ 5t^2 - 0.75t + 2.09 = 0$$
$$t = \frac{0.75 \pm \sqrt{0.75^2 - 4\left(0.007\ 5\right)2.09}}{0.015} = 2.88\ \text{s}\ \text{ or }\ 97.1\ \text{s}$$

The first time the dog reaches the bone is $\boxed{2.88\ \text{s.}}$

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(b) If the dog passes the bone, he must run around the merry-go-round again. The dog will draw even with the bone when

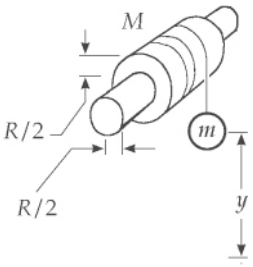
$$0.75t = \frac{2\pi}{3} + 2\pi + 0.007\,5t^2 \cdot$$

Solving this equation, we find (suppressing units)

$$t = \frac{0.75 \pm \sqrt{0.75^2 - 4(0.007\,5)8.38}}{0.015} = 12.8 \text{ s} \quad \text{or} \quad 87.2 \text{ s}$$

The dog draws even with the bone again at the time of $\boxed{12.8 \text{ s.}}$

P10.94 τ_f will oppose the torque due to the hanging object:



ANS. FIG. P10.94

$$\sum \tau = I\alpha = TR - \tau_f: \quad \tau_f = TR - I\alpha \qquad \textbf{[1]}$$

Now find T, I, and α in given or known terms and substitute into equation [1].

$$\sum F_y = T - mg = -ma: \quad T = m(g - a) \qquad \textbf{[2]}$$

also, $$\Delta y = v_i t + \frac{at^2}{2} a = \frac{2y}{t^2} \qquad \textbf{[3]}$$

and $$\alpha = \frac{a}{R} = \frac{2y}{Rt^2} \qquad \textbf{[4]}$$

with $$I = \frac{1}{2} M \left[R^2 + \left(\frac{R}{2} \right)^2 \right] = \frac{5}{8} MR^2 \qquad \textbf{[5]}$$

Substituting [2], [3], [4], and [5] into [1], we find

$$\tau_f = m\left(g - \frac{2y}{t^2} \right) R - \frac{5}{8} \frac{MR^2(2y)}{Rt^2} = \boxed{R\left[m\left(g - \frac{2y}{t^2} \right) - \frac{5}{4} \frac{My}{t^2} \right]}$$

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P10.2 (a) 0.209 rad/s^2; (b) yes

P10.4 144 rad

P10.6 -2.26×10^2 rad/s^2

P10.8 (a) 3.5 rad; (b) increase by a factor of 4

P10.10 Because the disk's average angular speed does not match the average angular speed expressed as $\left(\omega_i + \omega_f\right)/2$ in the model of a rigid object under constant angular acceleration, the angular acceleration of the disk cannot be constant.

P10.12 50.0 rev

P10.14 (a) $\boxed{\omega h^{3/2}\left(\frac{2}{g}\right)^{1/2}}$; (b) 1.16 cm; (c) The deflection is only 0.02% of the original height, so it is negligible in many practical cases; (d) Decrease

P10.16 ~10^7 rev/yr

P10.18 (a) 0.605 m/s; (b) 17.3 rad/s; (c) 5.82 m/s; (d) We did not need to know the length of the pedal cranks.

P10.20 (a) 54.3 rev; (b) 12.1 rev/s

P10.22 (a) 5.77 cm; (b) Yes. See P10.20 for full explanation.

P10.24 $\frac{a}{g}\sqrt{1+\pi^2}$

P10.26 (a) $\left(-2.73\hat{\mathbf{i}} + 1.24\hat{\mathbf{j}}\right)$ m; (b) It is in the second quadrant, at 156°;
(c) $\left(-1.85\hat{\mathbf{i}} - 4.10\hat{\mathbf{j}}\right)$ m/s; (d) It is moving toward the third quadrant, at 246°; (e) $\left(6.15\hat{\mathbf{i}} - 2.78\hat{\mathbf{j}}\right)$ m/s^2; (f) See ANS. FIG. P10.26;
(g) $\left(24.6\hat{\mathbf{i}} - 11.1\hat{\mathbf{j}}\right)$ N

P10.28 168 N · m

P10.30 (a) 1.03 s; (b) 10.3 rev

P10.32 (a) See ANS. FIG. P10.32; (b) 0.309 m/s^2; (c) T_1 = 7.67 N, T_2 = 9.22 N

P10.34 (a) For F = 25.1 N, R = 1.00 m. For F = 10.0 N, R = 25.1 m; (b) No. Infinitely many pairs of values that satisfy this requirement may exist: for any $F \le 50.0$ N, $R = 25.1$ N · m/F, as long as $R \le 3.00$ m.

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P10.36 (a) 1.95 s; (b) If the pulley were massless, the acceleration would be larger by a factor 35/32.5 and the time short by the square root of the factor 32.5/35. That is, the time would be reduced by 3.64%.

P10.38 $10^0\ \text{kg} \cdot \text{m}^2 = 1\ \text{kg} \cdot \text{m}^2$

P10.40 (a) See P10.40(a) for full description; (b) See P10.40(b) for full description

P10.42 $I_{y'} = \int_{\text{all mass}} r^2 dm = \int_0^L x^2 \frac{M}{L} dx = \frac{M}{L}\frac{x^3}{3}\bigg|_0^L = \frac{1}{3}ML^2$

P10.44 (a) 92.0 kg·m^2; (b) 184 J; (c) 6.00 m/s, 4.00 m/s, 8.00 m/s; (d) 184 J; (e) The kinetic energies computed in parts (b) and (d) are the same.

P10.46 $\frac{13}{24}MR^2\omega^2$

P10.48 276 J

P10.50 $v = \sqrt{\frac{2(m_1 - m_2)gh}{m_1 + m_2 + \frac{I}{R^2}}}$ and $\omega = \sqrt{\frac{2(m_1 - m_2)gh}{m_1R^2 + m_2R^2 + I}}$

P10.52 The situation is impossible because the range is only 3.86 km, not city-wide.

P10.54 (a) 6.90 J; (b) 8.73 rad/s; (c) 2.44 m/s; (d) The speed it attains in swinging is greater by 1.043 2 times

P10.56 $mr^2\left(\frac{2gh}{v^2} - 1\right)$

P10.58 (a) 74.3 W; (b) 401 W

P10.60 rolling: $v_f = \sqrt{10gh/7}$; sliding: $v_f = \sqrt{2gh}$; The time to roll is longer by a factor of $(0.7/0.5)^{1/2} = 1.18$

P10.62 (a) the cylinder; (b) $v^2/4g\sin\theta$; (c) The cylinder does not lose mechanical energy because static friction does not work on it. Its rotation means that it has 50% more kinetic energy than the cube at the start, and so it travels 50% farther up the incline.

P10.64 (a) 2.38 m/s; (b) The centripetal acceleration at the top is $\frac{v_2^2}{r} = \frac{(2.38\ \text{m/s})^2}{0.450\ \text{m}} = 12.6\ \text{m/s}^2 > g$. Thus, the ball must be in contact with the track, with the track pushing downward on it; (c) 4.31 m/s; (d) $\sqrt{-1.40\,\text{m}^2/\text{s}^2}$; (e) never makes it to the top of the loop

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P10.66 $\frac{1}{3}$ the length of the chimney

P10.68 (a) $d = (1890 + 80n)\left(\frac{0.459\,\text{m}}{80n - 150}\right)$; (b) 94.1 m; (c) 1.62 m; (d) –5.79 m; (e) The rising car will coast to a stop only for $n \geq 2$; (f) For $n = 0$ or $n = 1$, the mass of the elevator is less than the counterweight, so the car would accelerate upward if released; (g) 0.459 m

P10.70 $\omega(t) = \omega + At + \frac{1}{2}Bt^2$; (b) $\omega t + \frac{1}{2}At^2 + \frac{1}{6}Bt^3$

P10.72 (a) (i) –794 N · m, (ii) –2 510 N · m, (iii) 0 N · m, (iv) –1 160 N · m, (v) 2 940 N · m; (b) See P10.72(b) for full description.

P10.74 $-0.322\ \text{rad/s}^2$

P10.76 (a) 2.57×10^{29} J; (b) -1.63×10^{17} J/day

P10.78 (a) $Mg/3$; (b) $2g/3$; (c) $\sqrt{4gh/3}$; (d) The answer is the same.

P10.80 (a) $\theta \leq 35.5°$; (b) 0.184 m from the moving end

P10.82 (a) $a_{\text{CM}} = \frac{4F}{3M}$; (b) $\frac{1}{3}F$; (c) $\sqrt{\frac{8Fd}{3M}}$

P10.84 (a) 35.0 m/s^2; (b) $7.35\hat{\mathbf{i}}$ N; (c) 17.5 m/s^2; (d) $-3.68\hat{\mathbf{i}}$ N; (e) 0.827 m (from the top)

P10.86 54.0°

P10.88 See P10.88 for full design and specifications of flywheel.

P10.90 (a) See P10.90(a) for full solution; (b) See P10.90(g) for full solution; (c) $\frac{2\pi r_i}{h}\left(\sqrt{1 + \frac{vh}{\pi r_i^2}t} - 1\right)$; (d) $\alpha = -\frac{hv^2}{2\pi r_i^2\left(1 + \frac{vh}{\pi r_i^2}t\right)^{3/2}}$

P10.92 (a) See P10.92(a) for full explanation; (b) $\frac{2Mg(\sin\theta - \mu\cos\theta)}{2M + m}$

P10.94 $R\left[m\left(g - \frac{2y}{t^2}\right) - \frac{5}{4}\frac{My}{t^2}\right]$

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